

# Algebraic Geometry

Lecture Notes

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# Chapter 1

## Kähler Differentials

### 1.1 Review and Basic Definitions

Let  $f: X \rightarrow Y$  be projective and let  $\mathcal{F} \in \text{Coh}(X)$  be flat over  $Y$ . Then

$$y \mapsto h^i(X_y, \mathcal{F}|_{X_y}) \quad \text{is upper semicontinuous,}$$

whereas

$$y \mapsto \chi(X_y, \mathcal{F}|_{X_y}) \quad \text{is locally constant.}$$

If  $f$  is also smooth over a field  $k$  of characteristic zero, then

$$h^{p,q}(X_y) = \dim H^q(X_y, \Omega_{X_y}^p)$$

is constant.

#### 1.1.1 Derivations

Let  $B$  be an  $A$ -algebra and let  $M$  be a  $B$ -module.

**Definition 1.1.** An  $A$ -derivation is an  $A$ -linear map  $d: B \rightarrow M$  satisfying the Leibniz rule

$$d(b_1 b_2) = b_1 d(b_2) + b_2 d(b_1).$$

The  $B$ -module of  $A$ -derivations is denoted by  $\text{Der}_A(B, M)$ .

*Remark 1.2.* Every  $A$ -derivation  $A \rightarrow B \xrightarrow{d} M$  vanishes on  $A$ ; in particular,  $d(1_B) = 0$ .

*Proof.* For  $a \in A$ ,  $A$ -linearity and the Leibniz rule give

$$d(a 1_B) = a d(1_B), \quad d(a 1_B) = a d(1_B) + d(a) 1_B.$$

Moreover,  $d(1_B) = d(1_B^2) = 2d(1_B)$ , so  $d(1_B) = 0$ , and hence  $d(a) = 0$ . □

**Definition 1.3.** The *module of Kähler differentials* of  $B$  over  $A$  is a  $B$ -module  $\Omega_{B/A}$ , together with an  $A$ -derivation  $d: B \rightarrow \Omega_{B/A}$ , having the following universal property: for every  $A$ -

derivation  $\varphi: B \rightarrow M$ , there is a unique  $B$ -linear map making the diagram commute:

$$\begin{array}{ccc} B & \xrightarrow{\varphi} & M \\ d \downarrow & \nearrow \exists! & \\ \Omega_{B/A} & & \end{array}$$

Equivalently,

$$\boxed{\mathrm{Der}_A(B, M) \cong \mathrm{Hom}_B(\Omega_{B/A}, M)}$$

naturally in  $M$ .

**Proposition 1.4.** *The pair  $(\Omega_{B/A}, d)$  exists.*

*Proof.* Take the free  $B$ -module on symbols  $db, b \in B$ , and quotient by the submodule generated by

$$d(a) = 0 \quad (a \in A), \quad d(b_1 b_2) = b_1 d(b_2) + b_2 d(b_1).$$

The map  $b \mapsto db$  is the universal  $A$ -derivation. □

## 1.2 Examples and the First Exact Sequence

*Remark 1.5.* For a morphism  $f: X \rightarrow Y$ , there is a quasicoherent sheaf  $\Omega_{X/Y}$  on  $X$ . The map

$$d: \mathcal{O}_X \longrightarrow \Omega_{X/Y}$$

is not  $\mathcal{O}_X$ -linear; it is an  $f^{-1}\mathcal{O}_Y$ -derivation.

**Example 1.6** (Polynomial rings). For  $B = A[x_1, \dots, x_n]$ ,

$$\Omega_{B/A} \cong B^{\oplus n} \cong \bigoplus_{i=1}^n B dx_i.$$

Indeed, the  $A$ -derivation

$$f \longmapsto \left( \frac{\partial f}{\partial x_i} \right)_{i=1}^n$$

induces one direction, and the inverse sends the  $i$ -th basis vector to  $dx_i$ . Geometrically,

$$\boxed{\Omega_{\mathbb{A}_A^n / \mathrm{Spec} A} \cong \mathcal{O}_{\mathbb{A}_A^n}^{\oplus n}.$$

**Example 1.7** (Localization). For a multiplicatively closed subset  $S \subset A$ ,  $\Omega_{S^{-1}A/A} = 0$ . Indeed,

$$0 = d(a) = d\left(s \frac{a}{s}\right) = s d\left(\frac{a}{s}\right) + \frac{a}{s} d(s) = s d\left(\frac{a}{s}\right),$$

and  $s$  is a unit in  $S^{-1}A$ .

**Example 1.8** (Quotients). If  $B = A/\mathfrak{a}$ , then  $\Omega_{B/A} = 0$ , since the composite

$$A \twoheadrightarrow B \xrightarrow{d} \Omega_{B/A}$$

is zero and the first map is surjective.

### 1.2.1 Two Natural Maps

For  $A \rightarrow B \rightarrow C$ , there are natural  $C$ -linear maps

$$\Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A}, \quad \Omega_{C/A} \longrightarrow \Omega_{C/B}.$$

The first is induced by  $B \rightarrow C \xrightarrow{d_{C/A}} \Omega_{C/A}$  and the adjunction

$$\mathrm{Hom}_B(\Omega_{B/A}, \Omega_{C/A}) \cong \mathrm{Hom}_C(\Omega_{B/A} \otimes_B C, \Omega_{C/A}).$$

The second follows because  $d_{C/B}$  is also an  $A$ -derivation.

**Proposition 1.9** (First fundamental exact sequence). *For  $A \rightarrow B \rightarrow C$ , the sequence*

$$\Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0$$

*is exact.*

Geometrically, for  $X \xrightarrow{f} Y \rightarrow Z$ , this becomes

$$f^* \Omega_{Y/Z} \longrightarrow \Omega_{X/Z} \longrightarrow \Omega_{X/Y} \longrightarrow 0.$$

The first arrow need not be injective.

*Proof.* Surjectivity follows directly from the presentations

$$\Omega_{C/A} = \frac{\langle dc : c \in C \rangle}{\langle d(a) = 0, \text{ Leibniz relations} \rangle}, \quad \Omega_{C/B} = \frac{\langle dc : c \in C \rangle}{\langle d(b) = 0, \text{ Leibniz relations} \rangle}.$$

For exactness in the middle, apply  $\mathrm{Hom}_C(-, M)$  for every  $C$ -module  $M$ . The resulting sequence is

$$\mathrm{Der}_B(C, M) \longrightarrow \mathrm{Der}_A(C, M) \longrightarrow \mathrm{Der}_A(B, {}_B M).$$

The kernel of restriction to  $B$  consists precisely of the  $B$ -derivations. □

**Example 1.10** (Failure of injectivity). For

$$k \longrightarrow k[x] \longrightarrow k[x]/(x) = k,$$

$\Omega_{k[x]/k} = k[x] dx$ , whereas  $\Omega_{k/k} = 0$ . For  $C = k[x]/(x^2)$ , the element  $x dx$  maps to zero in  $\Omega_{C/k}$ , since  $d(x^2) = 2x dx = 0$ .

**Example 1.11** (Polynomial extension). For  $C = B[x_1, \dots, x_n]$ , the sequence

$$0 \longrightarrow \Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0$$

splits because  $\Omega_{C/B} \cong C^{\oplus n}$  is free. Hence

$$\Omega_{C/A} \cong (\Omega_{B/A} \otimes_B C) \oplus \Omega_{C/B}.$$

For  $f = \sum_I a_I x^I$ , the coefficientwise differential

$$d_{B/A}(f) = \sum_I d_{B/A}(a_I) x^I$$

together with the polynomial differential gives a splitting.

### 1.3 Localization, Base Change, and the Conormal Sequence

**Corollary 1.12** (Localization). *For  $A \rightarrow B$  and  $C = S^{-1}B$ ,*

$$\boxed{S^{-1}\Omega_{B/A} \cong \Omega_{S^{-1}B/A}.$$

*Proof.* In the first fundamental sequence

$$\Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0,$$

the left term is  $S^{-1}\Omega_{B/A}$ , while  $\Omega_{S^{-1}B/B} = 0$ . The resulting surjection has inverse induced by

$$d\left(\frac{b}{s}\right) = \frac{s db - b ds}{s^2}.$$

□

For an open subset  $U \subseteq X$ ,

$$\Omega_{X/Y}|_U \cong \Omega_{U/Y}.$$

**Proposition 1.13** (Base change). *Let  $A \rightarrow B$  and  $A \rightarrow A'$ , and put  $B' = A' \otimes_A B$ . Then*

$$\boxed{\Omega_{B/A} \otimes_B B' \cong \Omega_{B'/A'}.$$

*In particular,*

$$\Omega_{B/A} \otimes_B B' \cong \Omega_{B/A} \otimes_A A'.$$

Geometrically, for a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array} \quad \lrcorner$$

one has  $\Omega_{X'/Y'} \cong g^* \Omega_{X/Y}$ .

*Proof.* The map

$$B' = B \otimes_A A' \xrightarrow{d_{B/A} \otimes 1} \Omega_{B/A} \otimes_A A'$$

is an  $A'$ -derivation. Conversely,

$$B \longrightarrow B' \xrightarrow{d_{B'/A'}} \Omega_{B'/A'}$$

is an  $A$ -derivation. The induced maps are inverse by universality. □



**Proposition 1.14** (Conormal sequence). *Let  $A \rightarrow B \rightarrow C = B/\mathfrak{b}$ . Then  $\Omega_{C/B} = 0$ , and*

$$\boxed{\mathfrak{b}/\mathfrak{b}^2 \xrightarrow{\delta} \Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow 0}$$

*is exact, where  $\delta(\bar{b}) = db \otimes 1$ .*

For a closed immersion  $X = V(\mathcal{J}) \hookrightarrow Y$  over  $Z$ ,

$$\mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega_{Y/Z}|_X \longrightarrow \Omega_{X/Z} \longrightarrow 0.$$

*Proof.* The map is well defined because, for  $b_1, b_2 \in \mathfrak{b}$ ,

$$d(b_1 b_2) \otimes 1 = (b_1 d(b_2) + b_2 d(b_1)) \otimes 1 = 0.$$

The first fundamental sequence gives the right-exact part. For a  $C$ -module  $M$ , its dual is

$$0 \rightarrow \mathrm{Der}_A(C, M) \rightarrow \mathrm{Der}_A(B, {}_B M) \rightarrow \mathrm{Hom}_B(\mathfrak{b}, {}_B M).$$

This is exact because an  $A$ -derivation  $B \rightarrow M$  descends to  $B/\mathfrak{b}$  exactly when it vanishes on  $\mathfrak{b}$ .  $\square$

**Corollary 1.15** (Finite generation). *If  $C$  is a finitely generated  $A$ -algebra, then  $\Omega_{C/A}$  is a finitely generated  $C$ -module.*

*Proof.* Write  $C = A[x_1, \dots, x_n]/\mathfrak{b}$ . The conormal sequence gives

$$C^{\oplus n} \twoheadrightarrow \Omega_{C/A}.$$

$\square$

Geometrically, if  $f: X \rightarrow Y$  is locally of finite type and  $X$  is locally Noetherian, then  $\Omega_{X/Y} \in \mathrm{Coh}(X)$ .

**Example 1.16** (Hypersurfaces). For  $C = A[x_1, \dots, x_n]/(f)$ ,

$$\Omega_{C/A} = \mathrm{coker} \left( (f)/(f^2) \xrightarrow{\delta} \Omega_{A[x_1, \dots, x_n]/A} \otimes C \right) = \frac{\bigoplus_i C dx_i}{(df)}.$$

Thus, if  $X = V(f) \subset \mathbb{A}_A^n$ ,

$$\Omega_{X/\mathrm{Spec} A} \cong \Omega_{\mathbb{A}_A^n/A}|_X / (df).$$

### 1.3.1 Cotangent and Tangent Spaces

**Corollary 1.17.** *Let  $A$  be a  $k$ -algebra and  $x = \mathfrak{m} \in \mathrm{Specm}(A)$  a  $k$ -rational point, so  $k(x) = A/\mathfrak{m} \cong k$ . Then*

$$\boxed{\Omega_{A/k} \otimes_A k(x) \cong \mathfrak{m}/\mathfrak{m}^2.}$$

*The dual  $\mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k)$  is the Zariski tangent space at  $x$ .*

The statement also holds at a non-rational point when  $k(x)/k$  is separable.

## 1.4 A Nodal Curve and the Cotangent Space

**Example 1.18** (The coordinate axes). Let  $B = k[x, y]$ ,  $\mathfrak{b} = (xy)$ , and  $C = k[x, y]/(xy)$ . Then

$$\mathfrak{b}/\mathfrak{b}^2 \xrightarrow{\delta} \Omega_{k[x,y]/k} \otimes C \longrightarrow \Omega_{C/k} \longrightarrow 0.$$

Here

$$\mathfrak{b}/\mathfrak{b}^2 = \frac{(xy)}{(x^2y^2)} \cong C, \quad \Omega_{k[x,y]/k} \otimes C \cong C^{\oplus 2},$$

and

$$xy \mapsto d(xy) = x dy + y dx \mapsto (y, x).$$

The map

$$C^{\oplus 2} \longrightarrow (x, y) \subseteq C, \quad (a, b) \mapsto ax - by$$

identifies  $\Omega_{C/k} \cong (x, y)$ . At the origin,

$$\Omega_{C/k} \otimes_C k \cong k\bar{x} \oplus k\bar{y}.$$

Thus  $\Omega_{C/k}$  is not free: its fiber has dimension two at the origin and dimension one elsewhere on  $V(xy)$ .

Other examples include the plane curves defined by

$$xy + y^3 + x^3, \quad xy + x^2 - y^2, \quad x^2 - y^3.$$

In the hypersurface case, the conormal sequence is short exact:

$$0 \longrightarrow \mathfrak{b}/\mathfrak{b}^2 \longrightarrow \Omega_{B/k} \otimes C \longrightarrow \Omega_{C/k} \longrightarrow 0.$$

Tensoring with  $k[x, y]/(x, y)$  yields

$$\mathrm{Tor}_1^C(\Omega_{C/k}, k) \cong k \xrightarrow{0} k^2 \xrightarrow{\sim} k^2 \longrightarrow 0.$$

### 1.4.1 Injectivity at a Rational Point

For  $k \rightarrow A \rightarrow A/\mathfrak{m}$ , the conormal sequence reads

$$\mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega_{A/k} \otimes_A k \longrightarrow \Omega_{k(\mathfrak{m})/k} \longrightarrow 0.$$

The right term vanishes when  $k(\mathfrak{m}) = k$ , and more generally when  $k(\mathfrak{m})/k$  is separable.

Choose a presentation

$$\tilde{A} = k[x_1, \dots, x_n] \twoheadrightarrow A$$

with kernel  $\mathfrak{b}$ , and let  $\tilde{\mathfrak{m}}$  be the inverse image of  $\mathfrak{m}$ . There is a commutative diagram with exact rows:

$$\begin{array}{ccccccc} \mathfrak{b} & \longrightarrow & \tilde{\mathfrak{m}}/\tilde{\mathfrak{m}}^2 & \longrightarrow & \mathfrak{m}/\mathfrak{m}^2 & \longrightarrow & 0 \\ \downarrow & & \downarrow \tilde{\delta} & & \downarrow \delta & & \\ \mathfrak{b}/\mathfrak{b}^2 & \longrightarrow & \Omega_{\tilde{A}/k} \otimes k & \longrightarrow & \Omega_{A/k} \otimes k & \longrightarrow & 0. \end{array}$$

If  $\tilde{\mathfrak{m}} = (x_1 - a_1, \dots, x_n - a_n)$ , then

$$\tilde{\mathfrak{m}}/\tilde{\mathfrak{m}}^2 \cong \bigoplus_i k \overline{(x_i - a_i)}, \quad \Omega_{\tilde{A}/k} \otimes k \cong \bigoplus_i k dx_i,$$

and  $\tilde{\delta}(\overline{(x_i - a_i)}) = dx_i$ . Hence  $\tilde{\delta}$  is an isomorphism and  $\delta$  is injective.

## 1.5 The Diagonal Description

**Corollary 1.19.** *Let*

$$I = \ker(B \otimes_A B \xrightarrow{\mu} B), \quad \mu(b_1 \otimes b_2) = b_1 b_2.$$

*Then*

$$\boxed{\Omega_{B/A} \cong I/I^2}$$

*as  $B$ -modules, under  $db \mapsto 1 \otimes b - b \otimes 1$ . The  $B$ -module structure is induced by the first factor.*

*Proof.* Define

$$\delta: B \longrightarrow I/I^2, \quad b \longmapsto 1 \otimes b - b \otimes 1.$$

It vanishes on  $A$ , and

$$\delta(b_1 b_2) - b_1 \delta(b_2) - b_2 \delta(b_1) = (1 \otimes b_1 - b_1 \otimes 1)(1 \otimes b_2 - b_2 \otimes 1) \in I^2.$$

Thus it induces  $\Omega_{B/A} \rightarrow I/I^2$ . The map is surjective: if  $\sum_i b_i \otimes b'_i \in I$ , then  $\sum_i b_i b'_i = 0$ , and

$$\sum_i b_i \otimes b'_i = \sum_i b_i (1 \otimes b'_i - b'_i \otimes 1).$$

For the universal property, give  $B \oplus M$  the square-zero  $B$ -algebra structure

$$(b_1, m_1)(b_2, m_2) = (b_1 b_2, b_1 m_2 + b_2 m_1).$$

Given an  $A$ -derivation  $d: B \rightarrow M$ , define

$$\phi: B \otimes_A B \longrightarrow B \oplus M, \quad b_1 \otimes b_2 \longmapsto (b_1 b_2, b_1 d(b_2)).$$

Then  $\phi(I) \subseteq M$  and  $\phi(I^2) \subseteq M^2 = 0$ , so  $\phi$  induces  $I/I^2 \rightarrow M$ . Thus  $I/I^2$  has the universal property of  $\Omega_{B/A}$ .  $\square$

## 1.6 Field Extensions and the Sheaf of Differentials

**Proposition 1.20.** *Let  $K/k$  be a finitely generated field extension. Then*

$$\dim_K \Omega_{K/k} \geq \text{trdeg}(K/k),$$

*with equality if and only if the extension is separable.*

### 1.6.1 Global Definition

For  $f: X \rightarrow Y$ , take affine opens

$$U = \operatorname{Spec} B \subseteq X, \quad V = \operatorname{Spec} A \subseteq Y, \quad f(U) \subseteq V,$$

and define  $\Omega_{U/V} = \widetilde{\Omega_{B/A}}$ . The localization property lets these sheaves glue to  $\Omega_{X/Y}$ .

If  $f$  is separated, the diagonal  $\Delta: X \rightarrow X \times_Y X$  is a closed immersion. If  $\Delta(X) = V(\mathcal{J})$ , then

$$\Omega_{X/Y} := \mathcal{J}/\mathcal{J}^2 = \mathcal{J} \otimes_{\mathcal{O}_{X \times_Y X}} \mathcal{O}_{\Delta(X)}.$$

For  $X = \operatorname{Spec} B$ , this agrees with the affine construction. For a non-separated morphism, the diagonal is locally a closed immersion and the same definition works locally.

**Exercise 1.21.** For  $X \rightarrow Y \rightarrow Z$ , construct  $\Omega_{X/Z} \rightarrow \Omega_{X/Y}$  from the diagonal description.

**Example 1.22.** For  $X = \mathbb{A}_A^n \rightarrow \operatorname{Spec} A$ ,

$$\Omega_{X/A} = \Omega_{A[x_1, \dots, x_n]/A} = \mathcal{O}_{\mathbb{A}_A^n}^{\oplus n}.$$

## 1.7 The Euler Sequence

**Proposition 1.23** (Euler sequence). *Let  $X = \mathbb{P}_A^n \rightarrow \operatorname{Spec} A$ . There is an exact sequence*

$$0 \longrightarrow \Omega_{\mathbb{P}_A^n/A} \longrightarrow \mathcal{O}_{\mathbb{P}_A^n}(-1)^{\oplus(n+1)} \xrightarrow{(x_0, \dots, x_n)} \mathcal{O}_{\mathbb{P}_A^n} \longrightarrow 0.$$

*Proof.* Write  $X = \operatorname{Proj}(B)$ , where  $B = A[x_0, \dots, x_n]$ . The last map comes from

$$B(-1)^{\oplus(n+1)} \xrightarrow{(x_0, \dots, x_n)} B.$$

Its sheafification is surjective because  $\bigcap_i V_+(x_i) = \emptyset$ . Let  $M$  be its kernel.

On

$$U_0 = D_+(x_0) = \operatorname{Spec} A \left[ \frac{x_1}{x_0}, \dots, \frac{x_n}{x_0} \right],$$

the kernel of the degree-zero localized map is freely generated by

$$\frac{1}{x_0} e_j - \frac{x_j}{x_0^2} e_0, \quad 1 \leq j \leq n.$$

The isomorphism with differentials sends

$$d \left( \frac{x_j}{x_0} \right) \mapsto \frac{1}{x_0} e_j - \frac{x_j}{x_0^2} e_0.$$

The analogous maps on all standard opens agree on overlaps and glue to

$$\Omega_{\mathbb{P}_A^n/A} \xrightarrow{\sim} \ker \left( \mathcal{O}(-1)^{\oplus(n+1)} \xrightarrow{(x_0, \dots, x_n)} \mathcal{O} \right).$$

□

**Example 1.24** (The projective line). For  $n = 1$ , the Euler sequence gives

$$0 \longrightarrow \Omega_{\mathbb{P}^1_A/A} \longrightarrow \mathcal{O}(-1)^{\oplus 2} \longrightarrow \mathcal{O} \longrightarrow 0,$$

so taking determinants yields

$$\Omega_{\mathbb{P}^1_A/A} \cong \mathcal{O}(-2).$$

**Example 1.25** (Non-splitting for  $n > 1$ ). For  $n > 1$ ,  $\Omega_{\mathbb{P}^n_A/A}$  is not a direct sum of line bundles. Twisting the Euler sequence by  $\mathcal{O}(1)$  and taking global sections gives

$$0 \longrightarrow H^0(\Omega_{\mathbb{P}^n_A/A}(1)) \longrightarrow H^0(\mathcal{O}^{\oplus(n+1)}) \longrightarrow H^0(\mathcal{O}(1)).$$

The map

$$A^{n+1} \xrightarrow{\sim} \bigoplus_i A x_i$$

is an isomorphism, so  $H^0(\Omega_{\mathbb{P}^n_A/A}(1)) = 0$ . If  $\Omega_{\mathbb{P}^n_A/A} \cong \bigoplus_i \mathcal{O}(a_i)$ , then

$$a_i \leq -2 \implies \sum_i a_i \leq -2n.$$

On the other hand,

$$\det(\Omega_{\mathbb{P}^n_A/A}) \cong \mathcal{O}(-n-1),$$

so  $\sum_i a_i = -n-1 > -2n$  for  $n > 1$ , a contradiction.

## Chapter 2

# Smoothness

### 2.1 The Jacobian Matrix and Differentials

Let

$$\mathfrak{a} = (f_1, \dots, f_r) \subset k[x_1, \dots, x_n], \quad X = V(\mathfrak{a}) \subset \mathbb{A}_k^n.$$

For a  $k$ -rational point  $x \in X(k)$ , define the Jacobian matrix

$$J_x = \left( \frac{\partial f_j}{\partial x_i}(x) \right) \in M_{n \times r}(k).$$

More generally, for a point  $x = \mathfrak{p}$ , set

$$J_x \in M_{n \times r}(k(\mathfrak{p})), \quad J_x = \left( \frac{\partial f_j}{\partial x_i}(x) \right).$$

**Claim 2.1.** *For  $x \in X(k)$ , there is a natural isomorphism*

$$\boxed{T_{X,x}^* \cong \operatorname{coker} \left( k^{\oplus r} \xrightarrow{J_x} k^{\oplus n} \right)}.$$

*Proof.* Let  $\tilde{\mathfrak{m}}$  be the maximal ideal of  $x$  in  $k[x_1, \dots, x_n]$ , and let  $\mathfrak{m}$  be its image in  $k[X]$ . From

$$0 \longrightarrow \mathfrak{a} \longrightarrow \tilde{\mathfrak{m}} \longrightarrow \mathfrak{m} \longrightarrow 0$$

one obtains

$$0 \longrightarrow \frac{\mathfrak{a}}{\mathfrak{a} \cap \tilde{\mathfrak{m}}^2} \longrightarrow \frac{\tilde{\mathfrak{m}}}{\tilde{\mathfrak{m}}^2} \longrightarrow \frac{\mathfrak{m}}{\mathfrak{m}^2} \longrightarrow 0.$$

The middle term has basis given by the classes of  $x_1 - x_1(x), \dots, x_n - x_n(x)$ . The first map sends the class of  $f_j$  to

$$df_j(x) = \sum_{i=1}^n \frac{\partial f_j}{\partial x_i}(x) dx_i,$$

so it is represented by  $J_x$ . Since  $T_{X,x}^* = \mathfrak{m}/\mathfrak{m}^2$ , the result follows.  $\square$

### 2.1.1 The Cotangent Space and the Sheaf of Differentials

Let  $A = k[x_1, \dots, x_n]/\mathfrak{a}$  be a finitely generated  $k$ -algebra, and let  $x = \mathfrak{m}$  be a maximal ideal. The conormal sequence for  $k \rightarrow A \rightarrow k(x)$  gives

$$\frac{\mathfrak{m}}{\mathfrak{m}^2} \longrightarrow \Omega_{X/k} \otimes k(x) \longrightarrow \Omega_{k(x)/k} \longrightarrow 0.$$

Thus there is a natural map

$$T_{X,x}^* \longrightarrow \Omega_{X/k} \otimes k(x),$$

which is surjective when  $k(x)/k$  is separable.

*Remark 2.2.* For a quotient  $C = B/\mathfrak{a}$ , the general conormal sequence is

$$\mathfrak{a}/\mathfrak{a}^2 \longrightarrow \Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow 0.$$

**Proposition 2.3.** *If  $X = V(\mathfrak{a}) \subset \mathbb{A}_k^n$  is of finite type over  $k$ , then at every  $x \in X$ ,*

$$\boxed{\text{corank } J_x = \dim_{k(x)} (\Omega_{X/k} \otimes k(x)).}$$

*Proof.* Tensoring the conormal sequence with  $k(x)$  gives

$$\mathfrak{a}/\mathfrak{a}^2 \otimes k(x) \longrightarrow \Omega_{\mathbb{A}_k^n/k} \otimes k(x) \longrightarrow \Omega_{X/k} \otimes k(x) \longrightarrow 0.$$

The image of the first map is precisely  $\text{im } J_x \subset \bigoplus_{i=1}^n k(x) dx_i$ . □

#### Warning

The sequence used here is not the same as

$$0 \longrightarrow \mathfrak{a}/(\mathfrak{a} \cap \tilde{\mathfrak{m}}^2) \longrightarrow \tilde{\mathfrak{m}}/\tilde{\mathfrak{m}}^2 \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \longrightarrow 0.$$

**Proposition 2.4.** *Under the same hypotheses,*

$$\boxed{\text{corank } J_x \geq \dim T_{X,x}^*}.$$

*Equivalently,*

$$\dim_{k(x)} (\Omega_{X/k} \otimes k(x)) \geq \dim T_{X,x} = \dim T_{X,x}^*.$$

*Proof.* Put

$$r = \dim_{k(x)} (\Omega_{X/k} \otimes k(x)) = \text{corank } J_x.$$

Locally one can embed  $X$  in a  $k$ -scheme  $Z$  with  $\dim T_{Z,x} = r$ . A closed immersion  $X \hookrightarrow Z$  induces a surjection on cotangent spaces and hence an injection

$$T_{X,x} \hookrightarrow T_{Z,x}.$$

Therefore  $\dim T_{X,x} \leq r$ . □

**Proposition 2.5** (Separability criterion). *There is an isomorphism*

$$T_{X,x}^* \xrightarrow{\sim} \Omega_{X/k} \otimes k(x)$$

if and only if  $k(x)/k$  is separable.

*Proof.* If  $k(x)/k$  is separable, then  $\Omega_{k(x)/k} = 0$ , so the map from the cotangent space is surjective. On the other hand,

$$\dim T_{X,x}^* \leq \text{corank } J_x = \dim(\Omega_{X/k} \otimes k(x)),$$

so the surjection is an isomorphism. Conversely, if the map is an isomorphism, the conormal sequence gives  $\Omega_{k(x)/k} = 0$ , which is equivalent to separability for this finitely generated residue-field extension.  $\square$

## 2.2 Definition of Smoothness

**Definition 2.6.** Let  $X$  be of finite type over  $k$ . We say that  $X$  is *smooth of dimension  $d$  over  $k$*  if:

- (i) every irreducible component of  $X$  has dimension  $d$ ;
- (ii) locally, after writing

$$X = \text{Spec}(k[x_1, \dots, x_n]/\mathfrak{a}), \quad \mathfrak{a} = (f_1, \dots, f_r),$$

one has

$$\text{corank } J_x = d \quad \text{for every } x \in X.$$

*Remark 2.7.* In condition (ii), one may work componentwise with the prime ideal defining the relevant irreducible component.

The differential-geometric intuition is to regard  $X$  locally as a fiber of

$$F: \mathbb{A}^n \longrightarrow \mathbb{A}^r, \quad F = (f_1, \dots, f_r).$$

At  $x \in F^{-1}(0)$ ,

$$dF_x: T_{\mathbb{A}^n, x} \longrightarrow T_{\mathbb{A}^r, F(x)}, \quad T_{X, x} = \ker(dF_x)$$

at a regular value. Singularities arise at critical values.

*Remark 2.8.* The Jacobian condition is independent of the chosen affine coordinates and presentation, by [theorem 2.3](#).

**Proposition 2.9.** *To verify the Jacobian condition in the definition of smoothness, it is enough to verify it at closed points.*

*Proof.* The rank of the Jacobian is determined by its minors. A minor vanishes at  $\mathfrak{p}$  exactly when it belongs to  $\mathfrak{p}$ . Every prime ideal is contained in a maximal ideal; hence a minor that vanishes at  $\mathfrak{p}$  also vanishes at a closed point lying over  $\mathfrak{p}$ . Equivalently, the nonvanishing of a fixed minor is an open condition.  $\square$

*Remark 2.10.* Smoothness is a relative notion: the statement is about the structural morphism  $X \rightarrow \text{Spec } k$ .

**Lemma 2.11** (Base change). *For every field extension  $k'/k$ ,*

$$X/k \text{ is smooth} \iff X_{k'} := X \times_k k' \text{ is smooth over } k'.$$



In particular, this holds for  $k' = \bar{k}$ .

*Proof.* Dimension is unchanged by a field extension. The assertion is local and depends only on the Jacobian matrix. Closed points of  $X_{\bar{k}}$  correspond to pairs

$$(x, k(x) \hookrightarrow \bar{k}), \quad x \in X_{\text{cl}}.$$

Under this correspondence,  $J_x$  maps to  $J_y$  by scalar extension

$$M_{n \times r}(k(x)) \longrightarrow M_{n \times r}(\bar{k}).$$

The rank of a matrix is unchanged by extension of the ground field. □

## 2.3 Smoothness, Regularity, and Local Freeness

**Theorem 2.12.** *Let  $X$  be of finite type over  $k$ . Then*

$$X/k \text{ is smooth of dimension } d \iff \begin{array}{l} X \text{ is equidimensional of dimension } d, \\ \Omega_{X/k} \text{ is locally free of rank } d. \end{array}$$

*Proof.* By [theorem 2.3](#),

$$\text{corank } J_x = \dim_{k(x)}(\Omega_{X/k} \otimes k(x)).$$

Thus the Jacobian condition says exactly that every fiber of the coherent sheaf  $\Omega_{X/k}$  has dimension  $d$ . Nakayama's lemma, applied locally to a presentation of  $\Omega_{X/k}$ , shows that this constant fiber dimension is equivalent to local freeness of rank  $d$  under the stated equidimensionality hypothesis. □

**Corollary 2.13.** *If  $k$  is perfect, then*

$$X/k \text{ is smooth} \iff X \text{ is regular.}$$

*Proof.* Over a perfect field every residue-field extension  $k(x)/k$  at a closed point is separable. Hence

$$T_{X,x}^* \cong \Omega_{X/k} \otimes k(x).$$

For a smooth  $d$ -dimensional scheme,

$$\dim \mathcal{O}_{X,x} \leq \dim T_{X,x} \leq \dim(\Omega_{X/k} \otimes k(x)) = d.$$

At closed points on an equidimensional  $d$ -dimensional scheme, the outer dimensions agree, so the local ring is regular. The converse follows by reversing the argument and using the separability criterion. □

### Warning

Over a nonperfect field, regularity need not imply smoothness. The missing hypothesis is the separability of the residue-field extension.

**Corollary 2.14.** *A smooth scheme over a field is reduced and regular. Its irreducible components are disjoint, and hence it is locally irreducible.*

*Proof.* After base change to  $\bar{k}$ , the scheme is regular by [theorem 2.13](#). Regular local rings are domains, so the base change is reduced and locally irreducible. These properties descend to  $X$ . In particular, distinct irreducible components cannot meet.  $\square$

**Corollary 2.15.** *The smooth locus  $X_{\text{sm}} \subseteq X$  is open.*

*Proof.* Locally, the required rank of  $J_x$  is detected by the nonvanishing of minors of a fixed size. This is an open condition. Equivalently, it is the locus on which the coherent sheaf  $\Omega_{X/k}$  is locally free of the expected rank.  $\square$

## 2.4 Examples

**Example 2.16** (Affine space). Since

$$\Omega_{\mathbb{A}_k^n/k} \cong \mathcal{O}_{\mathbb{A}_k^n}^{\oplus n},$$

the affine space  $\mathbb{A}_k^n$  is smooth of dimension  $n$ .

**Example 2.17** (Hypersurfaces). Let  $X = V(f) \subset \mathbb{A}_k^n$ . Then

$$J_x = \left( \frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right),$$

and

$$\text{corank } J_x = n - 1 \iff \frac{\partial f}{\partial x_i}(x) \neq 0 \text{ for some } i.$$

Consequently,

$$V(f) \text{ is smooth} \iff f, \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \text{ have no common zero.}$$

**Example 2.18** (The effect of characteristic). Suppose  $\text{char } k = p > 0$  and consider

$$f = x^p - t \in k(t)[x].$$

The scheme  $V(f)$  is the spectrum of a field and is therefore regular, but  $f' = 0$ , so it is not smooth over  $k(t)$ . This is precisely the phenomenon caused by the fact that  $k(t)$  is not perfect.

## 2.5 Smooth Closed Subschemes

**Proposition 2.19.** *Let  $X$  be smooth over  $k$ , and let  $Y \hookrightarrow X$  be a closed subscheme with ideal sheaf  $\mathcal{I}$ . Put  $c = \text{codim}(Y, X)$ . Then*

$$Y \text{ is smooth}$$

*if and only if*

$$0 \longrightarrow \mathcal{I}/\mathcal{I}^2 \longrightarrow \Omega_{X/k}|_Y \longrightarrow \Omega_{Y/k} \longrightarrow 0$$

is exact and  $\mathcal{J}/\mathcal{J}^2$  is locally free of rank  $c$ .

*Proof.* Suppose first that  $Y$  is smooth. The sheaves  $\Omega_{X/k}|_Y$  and  $\Omega_{Y/k}$  are locally free of ranks  $\dim X$  and  $\dim Y$ , respectively. The conormal sequence is right exact. Locally, the surjection onto  $\Omega_{Y/k}$  splits, so its kernel is locally free of rank

$$\dim X - \dim Y = c.$$

Comparison with the conormal sequence shows that  $\mathcal{J}/\mathcal{J}^2$  maps isomorphically onto this kernel.

Conversely, suppose the displayed sequence is exact and  $\mathcal{J}/\mathcal{J}^2$  is locally free of rank  $c$ . Since  $\Omega_{X/k}|_Y$  is locally free of rank  $\dim X$ , the sequence splits locally and  $\Omega_{Y/k}$  is locally free of rank

$$\dim X - c = \dim Y.$$

By [theorem 2.12](#),  $Y$  is smooth. □

### 2.5.1 A Local Construction

The converse can also be seen more geometrically. Let

$$K = \operatorname{im} (\mathcal{J}/\mathcal{J}^2 \rightarrow \Omega_{X/k}|_Y).$$

Near a point of  $Y$ , choose sections  $s_1, \dots, s_c \in \mathcal{J}$  whose differentials freely generate  $K$ . Put

$$\mathcal{J}' = (s_1, \dots, s_c) \subseteq \mathcal{J}, \quad Y' = V(\mathcal{J}') \subseteq X.$$

Then  $Y \subseteq Y'$ , and the conormal sequence for  $Y'$  is

$$\mathcal{J}'/\mathcal{J}'^2 \rightarrow \Omega_{X/k}|_{Y'} \rightarrow \Omega_{Y'/k} \rightarrow 0.$$

After shrinking around  $Y$ , the sections  $ds_1, \dots, ds_c$  remain linearly independent. Therefore

$$\mathcal{J}'/\mathcal{J}'^2 \xrightarrow{\sim} \operatorname{im} (\mathcal{J}'/\mathcal{J}'^2 \rightarrow \Omega_{X/k}|_{Y'}),$$

and

$$0 \rightarrow \mathcal{J}'/\mathcal{J}'^2 \rightarrow \Omega_{X/k}|_{Y'} \rightarrow \Omega_{Y'/k} \rightarrow 0$$

is exact. Its restriction to  $Y$  shows that  $\Omega_{Y'/k}$  is locally free near  $Y$ . Hence  $Y'$  is smooth of dimension  $\dim Y$  near  $Y$ . Since a smooth scheme is locally integral, the closed inclusion  $Y \subseteq Y'$  of the same dimension must be an equality locally. This recovers  $Y = Y'$ .

## 2.6 Local Complete Intersections

**Definition 2.20.** Let  $X$  be smooth over  $k$ , and let  $Y \hookrightarrow X$  be a closed subscheme with ideal sheaf  $\mathcal{J}$ . We say that  $Y$  is a *local complete intersection*, abbreviated *l.c.i.*, if

$\mathcal{J} \text{ is locally generated by } c = \operatorname{codim}(Y, X) \text{ elements.}$

**Example 2.21.** If both  $X$  and  $Y$  are smooth over  $k$ , then  $Y \hookrightarrow X$  is a local complete intersection. Indeed, [theorem 2.19](#) gives

$$0 \longrightarrow \mathcal{I}/\mathcal{I}^2 \longrightarrow \Omega_{X/k}|_Y \longrightarrow \Omega_{Y/k} \longrightarrow 0,$$

where  $\mathcal{I}/\mathcal{I}^2$  is locally free of rank  $c = \text{codim}(Y, X)$ . Nakayama's lemma then gives  $c$  local generators for  $\mathcal{I}$ .

*Remark 2.22.* If  $Y \hookrightarrow X$  is a local complete intersection, then

$$\boxed{\mathcal{I}/\mathcal{I}^2 \text{ is locally free of rank } c = \text{codim}(Y, X).}$$

*Remark 2.23.* If  $Y \hookrightarrow X$  is integral and  $k(Y)/k$  is separable, then the conormal sequence

$$0 \longrightarrow \mathcal{I}/\mathcal{I}^2 \longrightarrow \Omega_{X/k}|_Y \longrightarrow \Omega_{Y/k} \longrightarrow 0$$

is exact at the generic point, and hence on a dense open subset of  $Y$ . In this situation  $\mathcal{I}/\mathcal{I}^2$  need not be locally free on all of  $Y$ .

## Chapter 3

# Flatness

### 3.1 Preliminaries: Conormal and Normal Bundles

Let  $Y \hookrightarrow X$  be a smooth closed subscheme of a smooth scheme, with ideal sheaf  $\mathcal{I}$ . Dualizing the conormal sequence

$$0 \longrightarrow \mathcal{I}/\mathcal{I}^2 \longrightarrow \Omega_X|_Y \longrightarrow \Omega_Y \longrightarrow 0$$

gives the normal sequence

$$0 \longrightarrow T_Y \longrightarrow T_X|_Y \longrightarrow N_{Y/X} \longrightarrow 0.$$

Taking determinants gives the adjunction formula

$$\omega_Y \cong \omega_X|_Y \otimes \det(N_{Y/X}) \cong \omega_X|_Y \otimes \det(\mathcal{I}/\mathcal{I}^2)^{-1}.$$

**Example 3.1** (Hypersurfaces in projective space). Since

$$\omega_{\mathbb{P}^n} \cong \mathcal{O}_{\mathbb{P}^n}(-n-1),$$

a smooth hypersurface  $Y \subset \mathbb{P}^n$  of degree  $d$  satisfies

$$\omega_Y \cong \mathcal{O}_{\mathbb{P}^n}(d-n-1)|_Y.$$

#### 3.1.1 The Hartshorne Conjecture

The Hartshorne conjecture predicts that a smooth closed subvariety  $X \subset \mathbb{P}^n$  of codimension

$$c < \frac{n}{3}$$

is a complete intersection:

$$X = V(f_1, \dots, f_c).$$

For example, in codimension two and sufficiently large ambient dimension, one expects  $X = H_1 \cap H_2$ . A related splitting problem asks when a rank-two vector bundle on  $\mathbb{P}^n$  must have the form

$$\mathcal{E} \cong \mathcal{O}(a) \oplus \mathcal{O}(b).$$

### 3.2 Algebraic Definition

**Definition 3.2.** Let  $A$  be a ring and  $M$  an  $A$ -module. The module  $M$  is *flat over  $A$*  if the functor

$$M \otimes_A -$$

is exact.

Tensor product is always right exact, so flatness is equivalent to the preservation of injections. Thus, for every short exact sequence

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0,$$

flatness of  $M$  means that

$$0 \longrightarrow M \otimes_A M_1 \longrightarrow M \otimes_A M_2 \longrightarrow M \otimes_A M_3 \longrightarrow 0$$

is exact.

**Proposition 3.3** (Ideal criterion). *An  $A$ -module  $M$  is flat if and only if, for every finitely generated ideal  $I \subset A$ , the natural map*

$$I \otimes_A M \longrightarrow M$$

*is injective.*

**Definition 3.4.** A ring homomorphism  $A \rightarrow B$  is *flat* if  $B$  is flat as an  $A$ -module.

### 3.3 Standard Properties

**Proposition 3.5.** *The following properties hold.*

1. *If  $A$  is local and  $M$  is a finite  $A$ -module, then*

$$M \text{ is flat over } A \iff M \text{ is free over } A.$$

2. *For every multiplicatively closed subset  $S \subset A$ , the localization map*

$$A \longrightarrow S^{-1}A$$

*is flat.*

3. *If  $M$  is flat over  $A$  and  $A \rightarrow B$  is any ring map, then  $M \otimes_A B$  is flat over  $B$ .*
4. *If  $A \rightarrow B$  is flat and  $N$  is flat over  $B$ , then  $N$  is flat over  $A$ .*
5. *Flatness is local:*

$$M \text{ is flat over } A \iff M_{\mathfrak{p}} \text{ is flat over } A_{\mathfrak{p}} \text{ for every } \mathfrak{p} \in \operatorname{Spec} A.$$

*Proof.* The first assertion is the finite flat equals finite free theorem over a local ring. For the second, localization preserves all finite limits and colimits of modules, and in particular injections.

For the third, if  $N_1 \hookrightarrow N_2$  is an injection of  $B$ -modules, then

$$N_i \otimes_B (B \otimes_A M) \cong N_i \otimes_A M,$$

so injectivity follows from the flatness of  $M$ . The fourth follows by composing the two exact tensor functors. The final assertion follows because exactness of a sequence of modules can be checked after localization at all prime ideals.  $\square$

**Proposition 3.6** (Quotient by a relative non-zero-divisor). *Let  $(A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$  be a local homomorphism of Noetherian local rings, and assume that  $B$  is flat over  $A$ . If  $b \in B$  acts as a non-zero-divisor on the closed fiber  $B/\mathfrak{m}B$ , then  $b$  is a non-zero-divisor on  $B$  and*

$$B/(b)$$

*is flat over  $A$ .*

**Proposition 3.7** (Base change). *Let  $A \rightarrow B$ ,  $A \rightarrow A'$ , and  $B' = A' \otimes_A B$ . If  $N$  is flat over  $B$ , then*

$$N \otimes_B B' \cong N \otimes_A A'$$

*is flat over  $A'$ . In particular, flat ring maps remain flat after arbitrary base change.*

**Proposition 3.8** (Local criterion for a ring map). *For a ring homomorphism  $A \rightarrow B$ , the following are equivalent:*

1.  $A \rightarrow B$  is flat;
2.  $A_{\mathfrak{p}} \rightarrow B \otimes_A A_{\mathfrak{p}}$  is flat for every  $\mathfrak{p} \in \operatorname{Spec} A$ ;
3.  $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}$  is flat for every  $\mathfrak{q} \in \operatorname{Spec} B$ , where  $\mathfrak{p} = \mathfrak{q} \cap A$ .

*It is enough to check the last condition at maximal ideals of  $B$ .*

### 3.4 Flat Morphisms of Schemes

**Definition 3.9.** Let  $f: X \rightarrow Y$  be a morphism and  $\mathcal{F} \in \operatorname{Qcoh}(X)$ . The sheaf  $\mathcal{F}$  is *flat over  $Y$*  if, for every  $x \in X$ , the stalk  $\mathcal{F}_x$  is flat over  $\mathcal{O}_{Y, f(x)}$ . The morphism  $f$  is *flat* if  $\mathcal{O}_X$  is flat over  $Y$ .

**Example 3.10** (The identity morphism). The identity  $X \rightarrow X$  is flat because every local ring  $\mathcal{O}_{X, x}$  is flat over itself. If  $X$  is locally Noetherian and  $\mathcal{F}$  is coherent, then

$$\mathcal{F} \text{ is flat over } X \iff \mathcal{F} \text{ is locally free.}$$

For a coherent sheaf, local freeness can be detected by the local constancy of

$$x \longmapsto \dim_{k(x)} (\mathcal{F}_x \otimes k(x)).$$

**Proposition 3.11** (Affine translation). *Let  $f: \operatorname{Spec} B \rightarrow \operatorname{Spec} A$  correspond to  $A \rightarrow B$ , and let  $N$  be a  $B$ -module. Then*

$$\tilde{N} \text{ is flat over } \operatorname{Spec} A \iff N \text{ is flat over } A.$$

### 3.4.1 Basic Examples and Stability Properties

1. Every open immersion is flat, since it is locally induced by a localization.
2. A closed immersion is generally not flat. Locally it is given by  $A \rightarrow A/I$ ; over a local ring, flatness of this finite module forces  $A/I$  to be free, which is highly restrictive.
3. In a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array} \quad \lrcorner$$

if  $\mathcal{F}$  is flat over  $Y$ , then  $g'^*\mathcal{F}$  is flat over  $Y'$ . In particular, if  $f$  is flat, then  $f'$  is flat.

4. If  $X \xrightarrow{f} Y \xrightarrow{g} Z$ , if  $\mathcal{F}$  is flat over  $Y$ , and if  $g$  is flat, then  $\mathcal{F}$  is flat over  $Z$ .

### 3.4.2 Non-flat Examples

**Example 3.12** (Normalization). Let  $X$  be an integral Noetherian scheme and let

$$\nu: \tilde{X} \longrightarrow X$$

be a finite normalization. Then  $\nu$  is flat only when it is an isomorphism. In particular, the normalization of a singular curve is typically not flat over the original curve.

**Example 3.13** (The cusp). Let

$$A = k[t^2, t^3] \subset B = k[t].$$

The finite map  $\text{Spec } B \rightarrow \text{Spec } A$  is the normalization of the cusp and is not flat. For example, if  $I = (t^2, t^3) \subset A$ , the element

$$t^3 \otimes 1 - t^2 \otimes t \in I \otimes_A B$$

maps to zero in  $B$ , exhibiting the failure of the ideal criterion.

**Example 3.14** (Blow-ups). The blow-up morphism

$$\text{Bl}_Z(X) \longrightarrow X$$

is generally not flat: over the center  $Z$ , the exceptional fiber has larger dimension than the general fiber.

## 3.5 Dimension of Fibers

**Theorem 3.15** (Dimension inequality and flat equality). *Let  $f: X \rightarrow Y$  be locally of finite type between Noetherian schemes, let  $x \in X$ , and put  $y = f(x)$ . Then*

$$\dim \mathcal{O}_{X_y, x} \geq \dim \mathcal{O}_{X, x} - \dim \mathcal{O}_{Y, y}.$$



If  $f$  is flat at  $x$ , equality holds:

$$\dim \mathcal{O}_{X_y, x} = \dim \mathcal{O}_{X, x} - \dim \mathcal{O}_{Y, y}.$$

The converse is false in general.

*Proof sketch.* Work locally with a flat local homomorphism  $(A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ , and induct on  $\dim A$ . The case  $\dim A = 0$  reduces to the case of a field. When  $\dim A \geq 1$ , after the usual reduction choose a non-zero-divisor  $t \in \mathfrak{m}$ . Flatness ensures that multiplication by  $t$  remains injective on  $B$ , so  $t$  is also a non-zero-divisor there. One compares the dimensions of  $A/(t)$ ,  $B/(t)$ , and their closed fiber, and then applies the induction hypothesis.  $\square$

**Corollary 3.16.** *A flat morphism has fibers of the expected dimension. If source and target are smooth and equidimensional, the fiber dimension is locally constant.*

### 3.6 Flatness over a Smooth Curve

**Proposition 3.17.** *Let  $f: X \rightarrow Y$  be a finite-type morphism, where  $Y$  is a smooth curve. Let  $x \in X$ , put  $y = f(x)$ , and let  $t \in \mathcal{O}_{Y, y}$  be a local parameter. Then  $f$  is flat at  $x$  if and only if multiplication by  $t$  on  $\mathcal{O}_{X, x}$  is injective.*

*Proof.* The local ring  $\mathcal{O}_{Y, y}$  is a discrete valuation ring. A module over a valuation domain is flat if and only if it is torsion-free. For the  $\mathcal{O}_{Y, y}$ -algebra  $\mathcal{O}_{X, x}$ , torsion-freeness is equivalent to injectivity of multiplication by a uniformizer  $t$ .  $\square$

**Corollary 3.18.** *Let  $X$  be reduced. A finite-type morphism  $X \rightarrow Y$  to a smooth curve is flat if every irreducible component of  $X$  dominates  $Y$ .*

*Proof.* For a reduced Noetherian ring, the zero-divisors are the union of the minimal primes. A local parameter pulled back from  $Y$  lies in a minimal prime exactly when the corresponding irreducible component is contained in a fiber. The hypothesis therefore makes multiplication by the parameter injective, and [theorem 3.17](#) applies.  $\square$

Geometrically, a vertical component appearing only in a special fiber is an obstruction to flatness.

### 3.7 Generic Flatness

**Theorem 3.19** (Generic flatness). *Let  $f: X \rightarrow Y$  be a finite-type morphism, let  $Y$  be integral and Noetherian, and let  $\mathcal{F}$  be a coherent sheaf on  $X$ . There exists a nonempty open subset  $U \subset Y$  such that*

$$\mathcal{F}|_{f^{-1}(U)}$$

*is flat over  $U$ .*

*Affine proof idea.* Write  $Y = \operatorname{Spec} A$ ,  $X = \operatorname{Spec} B$ , where  $A$  is a domain and  $B$  is a finite-type  $A$ -algebra, and let  $M$  be the finite  $B$ -module corresponding to  $\mathcal{F}$ . Generic freeness provides a nonzero  $a \in A$  such that  $M_a$  is a free, possibly infinite-rank,  $A_a$ -module. It is therefore

flat over  $A_a$ . A finite affine cover of  $X$  supplies a single element  $a$  that works on the whole inverse image of  $D(a)$ .  $\square$

### 3.8 Flatness and Cohomology

Let  $f: X \rightarrow Y$  be proper, and let  $\mathcal{F}$  be coherent and flat over  $Y$ . For  $y \in Y$ , write

$$X_y = f^{-1}(y), \quad \mathcal{F}_y = \mathcal{F}|_{X_y}.$$

**Theorem 3.20** (Upper semicontinuity). *The functions*

$$y \longmapsto h^i(y) := \dim_{k(y)} H^i(X_y, \mathcal{F}_y)$$

*are upper semicontinuous.*

There are natural base-change maps

$$(R^i f_* \mathcal{F}) \otimes k(y) \longrightarrow H^i(X_y, \mathcal{F}_y).$$

Under the hypotheses of the cohomology-and-base-change theorem, these maps are isomorphisms, and local freeness of  $R^i f_* \mathcal{F}$  is related to constancy of the cohomology dimensions of the fibers.

#### 3.8.1 Local Linear-algebra Model

Locally on  $Y = \operatorname{Spec} A$ , the relevant derived pushforward can be represented by a bounded complex of finite free modules, for example

$$A^p \xrightarrow{\alpha} A^q \xrightarrow{\beta} A^r.$$

After tensoring with  $k(y)$ , its cohomology computes the cohomology of the fiber. The ranks of the matrices  $\alpha(y)$  and  $\beta(y)$  vary semicontinuously, yielding upper semicontinuity of  $h^i(y)$ . If  $h^i(y)$  is locally constant and the adjacent base-change conditions hold, the corresponding higher direct image is locally free and commutes with base change.

### 3.9 Flat Morphisms Are Open

**Theorem 3.21.** *A flat morphism locally of finite presentation is an open map.*

*Idea of proof.* It is enough to show that the image of a basic open subset is open. In the affine case, write  $X = \operatorname{Spec} B$ ,  $Y = \operatorname{Spec} A$ , with  $B$  flat and finitely presented over  $A$ , and consider  $D(b) \subset X$ . The complement of its image consists of the points of  $\operatorname{Spec} A$  over which  $b$  vanishes identically on the fiber. Finite presentation, together with flatness, shows that this condition is closed. Hence the image of  $D(b)$  is open.  $\square$

### 3.10 Projective Morphisms and Hilbert Polynomials

Let  $f: X \rightarrow Y$  be projective, choose an embedding

$$X \hookrightarrow \mathbb{P}_Y^n,$$

and let  $\mathcal{F}$  be a coherent sheaf on  $X$ . For  $y \in Y$ , define

$$P_{\mathcal{F}_y}(m) = \chi(X_y, \mathcal{F}_y(m)).$$

For  $m \gg 0$ ,

$$P_{\mathcal{F}_y}(m) = \dim_{k(y)} H^0(X_y, \mathcal{F}_y(m)),$$

because the higher cohomology groups vanish.

**Theorem 3.22** (Flatness criterion via Hilbert polynomials). *Assume that  $Y$  is locally Noetherian. Then  $\mathcal{F}$  is flat over  $Y$  if and only if*

$$y \mapsto P_{\mathcal{F}_y}$$

*is locally constant. If  $Y$  is connected, this means that the Hilbert polynomial is independent of  $y$ .*

**Corollary 3.23.** *In a projective flat family, the fibers have the same Hilbert polynomial. Consequently, all numerical invariants encoded by that polynomial are constant, including dimension, degree, and arithmetic genus.*

**Remark 3.24** (Finite morphisms). For a finite morphism, the Hilbert polynomial is the length of the fiber. A finite, finitely presented morphism is flat precisely when the fiber length is locally constant.

### 3.11 Miracle Flatness

**Theorem 3.25** (Miracle flatness). *Let  $f: X \rightarrow Y$  be locally of finite type. Assume that:*

1.  *$X$  is Cohen–Macaulay;*
2.  *$Y$  is regular;*
3. *every fiber has the expected dimension*

$$\dim X - \dim Y.$$

*Then  $f$  is flat.*

A common local-algebra formulation is the following.

**Theorem 3.26.** *Let  $(A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$  be a local homomorphism of Noetherian local rings. Assume that  $A$  is regular,  $B$  is Cohen–Macaulay, and*

$$\dim B = \dim A + \dim(B/\mathfrak{m}B).$$

*Then  $B$  is flat over  $A$ .*

*Proof idea.* Choose a regular system of parameters in  $A$ . The dimension equality and the Cohen–Macaulay property show that their images form a regular sequence in  $B$ . Repeated application of the local criterion for flatness, starting with the residue field, then proves that  $B$  is flat over  $A$ .  $\square$

### 3.12 Smoothness from Flatness and Smooth Fibers

**Theorem 3.27.** *Let  $f: X \rightarrow Y$  be locally of finite presentation. Then  $f$  is smooth if and only if:*

1.  *$f$  is flat;*
2. *every geometric fiber of  $f$  is smooth.*

Thus smoothness separates into a global compatibility condition (flatness) and a fiber-wise regularity condition (geometric smoothness). For a projective family over a connected base, flatness also gives a constant Hilbert polynomial, after which smoothness may be studied fiber by fiber.

## Chapter 4

# Smooth Morphisms, Sard's Theorem, and Bertini's Theorem

### 4.1 Smooth Morphisms

Until now, smoothness has mainly been discussed for schemes over a field. We now consider a morphism  $f: X \rightarrow Y$  that is locally of finite presentation.

**Definition 4.1.** The morphism  $f$  is *smooth at*  $x \in X$  if:

1.  $f$  is flat at  $x$ ;
2. the fiber  $X_y$ , where  $y = f(x)$ , is smooth over  $k(y)$  at  $x$ .

The morphism is *smooth* if it is smooth at every point of  $X$ .

Equivalently, one may require the geometric fibers to be regular. The finite-presentation hypothesis is part of the definition and is essential in the standard structure theorems.

**Example 4.2** (Products). If  $Z$  is smooth over  $k$ , then the projection

$$Y \times_k Z \longrightarrow Y$$

is smooth. Indeed, it is the base change of  $Z \rightarrow \operatorname{Spec} k$ .

**Example 4.3** (Projective bundles). If  $\mathcal{E}$  is locally free of rank  $n + 1$  on  $Y$ , then

$$\mathbb{P}(\mathcal{E}) \longrightarrow Y$$

is smooth of relative dimension  $n$ . Locally on  $Y$ , it is the projection  $\mathbb{P}_Y^n \rightarrow Y$ .

*Remark 4.4.* Smoothness is local on both source and target and is preserved by arbitrary base change and by composition.

#### 4.1.1 The Differential Criterion

**Theorem 4.5.** Let  $f: X \rightarrow Y$  be locally of finite presentation, flat, and with fibers of pure dimension  $n$ . Then

|                                                                             |
|-----------------------------------------------------------------------------|
| $f \text{ is smooth} \iff \Omega_{X/Y} \text{ is locally free of rank } n.$ |
|-----------------------------------------------------------------------------|

*Proof.* Formation of Kähler differentials commutes with base change, so for every  $y \in Y$ ,

$$\Omega_{X/Y}|_{X_y} \cong \Omega_{X_y/k(y)}.$$

If  $f$  is smooth, every fiber is smooth of dimension  $n$ ; hence  $\Omega_{X_y/k(y)}$  is locally free of rank  $n$ . Flatness and the local criterion for finite modules then imply that  $\Omega_{X/Y}$  is locally free of rank  $n$ .

Conversely, if  $\Omega_{X/Y}$  is locally free of rank  $n$ , its restriction to every fiber is locally free of rank  $n$ . The fiberwise differential criterion from [theorem 2.12](#) shows that every fiber is smooth. Together with flatness, this proves that  $f$  is smooth.  $\square$

*Remark 4.6.* Local freeness of  $\Omega_{X/Y}$  alone does not imply smoothness: flatness and the expected fiber dimension are also needed. This point is especially important for nonreduced schemes and in positive characteristic.

## 4.2 The Local Standard Form

Smooth morphisms are locally projections from affine space, up to an étale morphism.

**Theorem 4.7** (Local structure of a smooth morphism). *Let  $f: X \rightarrow Y$  be smooth of relative dimension  $n$ , and let  $x \in X$ . After replacing  $X$  and  $Y$  by open neighborhoods of  $x$  and  $f(x)$ , respectively, there is an étale morphism*

$$g: X \rightarrow \mathbb{A}_Y^n$$

*such that  $f$  is the composite*

$$X \xrightarrow{g} \mathbb{A}_Y^n \rightarrow Y.$$

*Construction.* Choose local sections  $t_1, \dots, t_n \in \mathcal{O}_X$  such that

$$dt_1, \dots, dt_n$$

form a basis of  $\Omega_{X/Y}$  near  $x$ . They define a  $Y$ -morphism

$$g = (t_1, \dots, t_n): X \rightarrow \mathbb{A}_Y^n.$$

The exact sequence of differentials for  $X \rightarrow \mathbb{A}_Y^n \rightarrow Y$  gives

$$g^* \Omega_{\mathbb{A}_Y^n/Y} \rightarrow \Omega_{X/Y} \rightarrow \Omega_{X/\mathbb{A}_Y^n} \rightarrow 0.$$

The first map is an isomorphism near  $x$ , so  $\Omega_{X/\mathbb{A}_Y^n} = 0$  there. The map  $g$  is also flat near  $x$ , after shrinking, and hence is étale.  $\square$

### 4.2.1 Cutting Down to the Expected Relative Dimension

Suppose locally that  $X \hookrightarrow Z = \mathbb{A}_Y^d$  has ideal sheaf  $\mathcal{I}$ . The conormal sequence is

$$\mathcal{I}/\mathcal{I}^2 \rightarrow \Omega_{Z/Y}|_X \rightarrow \Omega_{X/Y} \rightarrow 0.$$

If  $\Omega_{X/Y}$  is locally free of rank  $n$ , then the image of the first arrow is locally free of rank  $d - n$ . One can choose

$$f_1, \dots, f_{d-n} \in \mathcal{I}_x$$

whose differentials form a basis of that image at  $x$ . After shrinking,

$$X \subseteq Z' := V(f_1, \dots, f_{d-n}) \subseteq Z,$$

and

$$\Omega_{Z'/Y} \cong \Omega_{Z/Y} / \langle df_1, \dots, df_{d-n} \rangle.$$

The relative non-zero-divisor criterion from [theorem 3.6](#) is used successively to preserve flatness while cutting down.

### 4.3 Generic Smoothness

**Theorem 4.8** (Generic smoothness). *Let  $f: X \rightarrow Y$  be a dominant morphism of integral schemes of finite type over a field  $k$ . Assume that the extension*

$$k(X)/k(Y)$$

*is separable. Then the smooth locus of  $f$  contains a nonempty open subset of  $X$ . More precisely, there are nonempty open subsets  $U \subseteq X$  and  $V \subseteq Y$ , with  $f(U) \subseteq V$ , such that*

$$f|_U: U \longrightarrow V$$

*is smooth.*

*Proof.* Separability gives

$$\dim_{k(X)} \Omega_{k(X)/k(Y)} = \text{trdeg}(k(X)/k(Y)) = \dim X - \dim Y =: n.$$

Since  $\Omega_{X/Y}$  is coherent and

$$\Omega_{X/Y} \otimes k(X) \cong \Omega_{k(X)/k(Y)},$$

there is a nonempty open subset on which  $\Omega_{X/Y}$  is locally free of rank  $n$ . By generic flatness, we may shrink further so that  $f$  is flat. The fiber-dimension formula then gives fibers of dimension  $n$ , and [theorem 4.5](#) shows that the resulting restriction of  $f$  is smooth.  $\square$

#### Warning

The separability hypothesis in generic smoothness cannot be omitted. Purely inseparable extensions in positive characteristic give dominant morphisms that are nowhere smooth.

**Corollary 4.9** (Characteristic zero). *Let  $X$  and  $Y$  be integral schemes of finite type over a field of characteristic zero, and let  $f: X \rightarrow Y$  be dominant. There is a nonempty open subset  $V \subseteq Y$  such that*

$$f^{-1}(V) \longrightarrow V$$

is smooth after removing from the source the closed complement of the smooth locus. If  $X$  and  $Y$  are smooth, one may choose  $V$  so that the entire restriction  $f^{-1}(V) \rightarrow V$  is smooth.

For a nondominant morphism, an open subset of  $Y$  over which the morphism is smooth may be empty on the source. For instance, if  $X \hookrightarrow Y$  is a proper closed immersion, then  $Y \setminus X$  has empty inverse image.

## 4.4 Relative Tangent Bundles

Let  $X$  and  $Y$  be smooth over  $k$ , and let  $f: X \rightarrow Y$  be smooth. The exact sequence of differentials is

$$0 \longrightarrow f^*\Omega_{Y/k} \longrightarrow \Omega_{X/k} \longrightarrow \Omega_{X/Y} \longrightarrow 0.$$

The left term has rank  $\dim Y$ , while the right term has rank  $\dim X - \dim Y$ . Dualizing gives

$$0 \longrightarrow T_{X/Y} \longrightarrow T_{X/k} \longrightarrow f^*T_{Y/k} \longrightarrow 0.$$

After restriction to a fiber  $X_y$ , this becomes the normal sequence

$$0 \longrightarrow T_{X_y/k(y)} \longrightarrow T_{X/k}|_{X_y} \longrightarrow f^*T_{Y/k}|_{X_y} \longrightarrow 0.$$

**Proposition 4.10** (Infinitesimal submersion criterion). *Let  $X$  and  $Y$  be smooth over  $k$ , and let  $f: X \rightarrow Y$  be locally of finite presentation. Then  $f$  is smooth at  $x$  if and only if the tangent map*

$$df_x: T_x X \longrightarrow T_{f(x)} Y \otimes_{k(f(x))} k(x)$$

*is surjective.*

*Proof.* The dual of  $df_x$  is the map on cotangent spaces induced by

$$f^*\Omega_{Y/k} \longrightarrow \Omega_{X/k}.$$

Surjectivity of  $df_x$  is equivalent to the expected rank of  $\Omega_{X/Y} \otimes k(x)$ . Since source and target are smooth, the Jacobian criterion and miracle flatness yield flatness at  $x$ ; the differential criterion then proves smoothness.  $\square$

## 4.5 Algebraic Sard Theorem

**Theorem 4.11** (Sard theorem in characteristic zero). *Let  $X$  and  $Y$  be smooth integral schemes of finite type over a field  $k$  of characteristic zero, and let  $f: X \rightarrow Y$ . For  $r \geq 0$ , put*

$$X_r = \{x \in X \mid \text{rank}(T_x X \rightarrow T_{f(x)} Y \otimes k(x)) \leq r\}.$$

*Then*

$$\dim \overline{f(X_r)} \leq r.$$

*Proof.* Let  $Y'$  be an irreducible component of  $\overline{f(X_r)}$ , and choose an irreducible subvariety  $X' \subseteq X_r$  that dominates  $Y'$ . In characteristic zero the extension  $k(X')/k(Y')$  is separable.



By generic smoothness, after replacing  $X'$  and  $Y'$  by nonempty open subsets, the induced map

$$X' \longrightarrow Y'$$

is smooth. At a point of this smooth locus, its tangent map is surjective onto  $T_{Y'}$ . Because  $X' \subseteq X_r$ , its rank is at most  $r$ . Hence

$$\dim Y' \leq r.$$

This holds for every component  $Y'$ , proving the theorem.  $\square$

**Corollary 4.12.** *Let  $f: X \rightarrow Y$  be a dominant morphism between smooth integral schemes of finite type over a field of characteristic zero. There is a nonempty open subset  $V \subseteq Y$  such that*

$$f^{-1}(V) \longrightarrow V$$

*is smooth.*

*Proof.* Take  $r = \dim Y - 1$ . Sard's theorem gives

$$\dim \overline{f(X_r)} < \dim Y.$$

Set

$$V = Y \setminus \overline{f(X_r)}.$$

For every  $x \in f^{-1}(V)$ , the tangent map  $df_x$  is surjective. The infinitesimal submersion criterion [theorem 4.10](#) now applies.  $\square$

## 4.6 Bertini's Theorem

Let  $X$  be a smooth quasi-projective variety over  $k$ , let  $\mathcal{L}$  be a line bundle on  $X$ , and let

$$W \subseteq H^0(X, \mathcal{L})$$

be a finite-dimensional basepoint-free linear system. Write

$$|W| = \mathbb{P}(W)$$

for its parameter space.

**Theorem 4.13** (Bertini). *Assume that  $k = \bar{k}$  has characteristic zero. There is a nonempty dense open subset  $U \subseteq |W|$  such that, for every  $[s] \in U$ ,*

$$V(s) \subseteq X$$

*is smooth.*

### 4.6.1 The Universal Hyperplane Construction

Define the incidence variety

$$\mathcal{H}_X = \{(x, [s]) \in X \times |W| \mid s(x) = 0\}.$$

It fits into the diagram

$$\begin{array}{ccc} \mathcal{H}_X & \hookrightarrow & X \times |W| \\ p \downarrow & \lrcorner & \downarrow \text{pr}_1 \\ X & \longrightarrow & X \end{array} \quad \begin{array}{ccc} \mathcal{H}_X & \hookrightarrow & X \times |W| \\ \pi \downarrow & & \downarrow \text{pr}_1 \\ |W| & \xlongequal{\quad} & |W|. \end{array}$$

The fiber of  $\pi$  over  $[s]$  is  $V(s)$ .

Because  $W$  is basepoint free, evaluation gives a surjection

$$W \otimes \mathcal{O}_X \longrightarrow \mathcal{L}.$$

Let  $\mathcal{Q}$  be its kernel. Then

$$\mathcal{H}_X \cong \mathbb{P}_X(\mathcal{Q}),$$

so  $p: \mathcal{H}_X \rightarrow X$  is a projective-space bundle. Consequently,  $\mathcal{H}_X$  is smooth because  $X$  is smooth.

*Proof of theorem 4.13.* Apply generic smoothness to

$$\pi: \mathcal{H}_X \longrightarrow |W|.$$

There is a nonempty open subset  $U \subseteq |W|$  over which  $\pi$  is smooth. For  $[s] \in U$ , its fiber

$$\pi^{-1}([s]) = V(s)$$

is therefore smooth. □

*Remark 4.14.* If  $X$  is projective, the set of parameters defining smooth members is open because the nonsmooth locus in the universal family has closed image under the proper projection to  $|W|$ . Without properness, the smooth parameters still contain a dense open subset, but the full set need not itself be open.

### 4.6.2 Positive Characteristic and Higher-degree Sections

For the complete hyperplane system of a projective embedding of a smooth variety over an algebraically closed field, a general hyperplane section is smooth in arbitrary characteristic. For a smaller linear system in positive characteristic, however, the associated morphism may be inseparable, and the characteristic-zero generic-smoothness proof can fail.

Sections of degree  $d$  can be treated using the Veronese embedding

$$X \subseteq \mathbb{P}^n \longrightarrow \mathbb{P}^N, \quad v_d.$$

The inverse image of a hyperplane under  $v_d$  is a degree- $d$  hypersurface in  $\mathbb{P}^n$ . Bertini applied

after the Veronese embedding therefore gives smooth general intersections with degree- $d$  hypersurfaces, subject to the same separability and basepoint conditions.

In arithmetic settings over finite fields, one can ask for the density of hypersurfaces  $H$  such that  $H \cap X$  is smooth. This leads to finite-field Bertini theorems, where the answer is expressed as an asymptotic density rather than as the existence of a  $k$ -rational point in a Zariski-open parameter set.

## 4.7 The Discriminant Divisor

Consider the parameter space

$$|\mathcal{O}_{\mathbb{P}^n}(d)| \cong \mathbb{P}^N$$

of degree- $d$  hypersurfaces, and define the discriminant

$$D = \{[s] \in \mathbb{P}^N \mid V(s) \text{ is singular}\}.$$

**Proposition 4.15.** *Assume  $d \geq 2$  and  $\text{char } k = 0$ . Then  $D$  is a proper irreducible hypersurface in  $\mathbb{P}^N$ . Its degree is*

$$\deg D = (n+1)(d-1)^n.$$

*There is a dense open subset  $D^\circ \subseteq D$  whose members have exactly one singular point, generically an ordinary double point.*

*Geometric description.* Introduce the incidence correspondence

$$\Sigma = \{(x, [s]) \in \mathbb{P}^n \times \mathbb{P}^N \mid s(x) = 0, ds(x) = 0\}.$$

For fixed  $x$ , these are linear conditions on  $s$ , so  $\Sigma$  is irreducible. Its image in  $\mathbb{P}^N$  is precisely  $D$ , proving irreducibility. A Chern-class computation gives the displayed degree. The locus of hypersurfaces with more than one singular point or with a worse singularity has higher codimension, which gives  $D^\circ$ .  $\square$

The singular locus of  $D$  records members with several singular points or with singularities worse than an ordinary double point. In this way,  $D$  organizes the degeneration types in the universal hypersurface family.

## Chapter 5

# Unramified and Étale Morphisms

Throughout this chapter, schemes are locally Noetherian and morphisms are locally of finite type unless otherwise stated.

### 5.1 Unramified Morphisms

**Definition 5.1.** Let  $f: X \rightarrow Y$ , let  $x \in X$ , and put  $y = f(x)$ . The morphism  $f$  is *unramified at  $x$*  if:

1.  $f$  is locally of finite type at  $x$ ;
- 2.

$$\mathfrak{m}_y \mathcal{O}_{X,x} = \mathfrak{m}_x;$$

3. the residue-field extension  $k(x)/k(y)$  is finite separable.

It is *unramified* if it is unramified at every point.

**Example 5.2** (Field extensions). For a finite extension  $K/k$ ,

$$\mathrm{Spec} K \longrightarrow \mathrm{Spec} k$$

is unramified if and only if  $K/k$  is separable.

**Example 5.3** (Number rings). Let  $K \subset L$  be a finite extension of number fields. For a prime  $\mathfrak{q} \subset \mathcal{O}_L$ , put  $\mathfrak{p} = \mathfrak{q} \cap \mathcal{O}_K$ . The morphism

$$\mathrm{Spec} \mathcal{O}_L \longrightarrow \mathrm{Spec} \mathcal{O}_K$$

is unramified at  $\mathfrak{q}$  precisely when

$$\mathfrak{p} \mathcal{O}_{L,\mathfrak{q}} = \mathfrak{q} \mathcal{O}_{L,\mathfrak{q}}$$

and the finite residue-field extension  $k(\mathfrak{q})/k(\mathfrak{p})$  is separable.

**Example 5.4** (A power map). Assume  $\mathrm{char} k \neq 2$ . The map

$$\mathbb{A}_k^1 \longrightarrow \mathbb{A}_k^1, \quad z \longmapsto z^2$$

is ramified at the origin and unramified away from it. At the origin, the image of the maximal ideal  $(t)$  is  $(z^2)$ , which does not generate  $(z)$ .

**Example 5.5.** Every open immersion and every closed immersion is unramified.

**Example 5.6** (Zero-dimensional schemes). Let

$$X = \operatorname{Spec}(k[t]/(F)), \quad F = \prod_i F_i^{n_i},$$

where the  $F_i$  are distinct irreducible polynomials. At the point defined by  $F_i$ , the morphism  $X \rightarrow \operatorname{Spec} k$  is unramified if and only if  $n_i = 1$  and  $F_i$  is separable.

### 5.1.1 Equivalent Criteria

**Theorem 5.7.** Let  $f: X \rightarrow Y$  be locally of finite type. The following are equivalent:

1.  $f$  is unramified;
2.  $\Omega_{X/Y} = 0$ ;
3. the diagonal

$$\Delta_{X/Y}: X \longrightarrow X \times_Y X$$

is an open immersion.

**Lemma 5.8** (Fiberwise criterion). The morphism  $f$  is unramified if and only if every fiber  $X_y$  is reduced and zero-dimensional and every extension  $k(x)/k(y)$ , for  $x \in X_y$ , is finite separable.

*Proof.* If  $f$  is unramified at  $x$ , then

$$\mathcal{O}_{X_y, x} = \mathcal{O}_{X, x} / \mathfrak{m}_y \mathcal{O}_{X, x} \cong k(x).$$

Thus the fiber is reduced and zero-dimensional near  $x$ , and the residue extension is separable.

Conversely, a reduced zero-dimensional local ring is a field. Hence  $\mathcal{O}_{X_y, x} = k(x)$ , which is equivalent to  $\mathfrak{m}_y \mathcal{O}_{X, x} = \mathfrak{m}_x$ . Together with separability, this is the definition of being unramified.  $\square$

*Proof of the differential criterion.* Formation of differentials commutes with base change:

$$\Omega_{X/Y}|_{X_y} \cong \Omega_{X_y/k(y)}.$$

If  $f$  is unramified, the fibers are locally spectra of finite separable extensions, so their modules of differentials vanish. Nakayama's lemma gives  $\Omega_{X/Y} = 0$ .

Conversely, suppose  $\Omega_{X/Y} = 0$ . On a fiber  $X_y = \operatorname{Spec} A$ , the conormal sequence at  $x$  gives

$$\mathfrak{m}_x / \mathfrak{m}_x^2 \longrightarrow \Omega_{A/k(y)} \otimes k(x) \longrightarrow \Omega_{k(x)/k(y)} \longrightarrow 0.$$

Both terms involving differentials vanish. Thus  $\mathfrak{m}_x / \mathfrak{m}_x^2 = 0$ , and Nakayama gives  $\mathfrak{m}_x = 0$  in  $\mathcal{O}_{X_y, x}$ ; moreover,  $\Omega_{k(x)/k(y)} = 0$ , so the residue extension is separable.  $\square$

### 5.1.2 Tangent Spaces

**Proposition 5.9.** *If  $f$  is unramified at  $x$ , then the tangent map*

$$df_x: T_{X,x} \longrightarrow T_{Y,y} \otimes_{k(y)} k(x)$$

*is injective.*

*Proof.* The exact sequence of differentials gives a surjection

$$f^*\Omega_Y \otimes k(x) \twoheadrightarrow \Omega_X \otimes k(x)$$

because  $\Omega_{X/Y} = 0$ . Dualizing over  $k(x)$  gives the asserted injection.  $\square$

**Example 5.10** (Normalization of a cusp). The normalization

$$\mathbb{A}^1 \longrightarrow V(x^3 - y^2), \quad t \longmapsto (t^2, t^3),$$

is ramified over the cusp: the induced map on tangent spaces at the origin is zero.

## 5.2 Étale Morphisms

**Definition 5.11.** A morphism  $f: X \rightarrow Y$  is *étale at  $x$*  if it is flat and unramified at  $x$ . It is *étale* if it is étale everywhere.

**Theorem 5.12.** *For a morphism  $f: X \rightarrow Y$  locally of finite presentation, the following are equivalent:*

1.  $f$  is étale;
2.  $f$  is flat and  $\Omega_{X/Y} = 0$ ;
3.  $f$  is smooth of relative dimension zero.

**Corollary 5.13.** *If  $f$  is étale at  $x$ , then*

$$df_x: T_{X,x} \xrightarrow{\sim} T_{Y,y} \otimes_{k(y)} k(x).$$

*Proof.* Unramifiedness gives injectivity. Flatness identifies the cotangent spaces through

$$\mathfrak{m}_y/\mathfrak{m}_y^2 \otimes_{k(y)} k(x) \xrightarrow{\sim} \mathfrak{m}_x/\mathfrak{m}_x^2,$$

so the tangent map is also surjective.  $\square$

**Proposition 5.14.** *Let  $X$  and  $Y$  be schemes over  $S$ . If  $f: X \rightarrow Y$  is étale, then*

$$\boxed{f^*\Omega_{Y/S} \xrightarrow{\sim} \Omega_{X/S}.}$$

*Proof.* The transitivity sequence is

$$f^*\Omega_{Y/S} \longrightarrow \Omega_{X/S} \longrightarrow \Omega_{X/Y} \longrightarrow 0.$$

The last term vanishes. Flatness of  $f$ , applied to the conormal description of differentials via diagonals, gives injectivity of the first arrow.  $\square$

### 5.3 The Ramification Divisor

Let  $X$  and  $Y$  be smooth integral  $k$ -schemes of the same dimension, and let  $f: X \rightarrow Y$  be dominant and generically finite. Assume  $k(X)/k(Y)$  is separable. Then

$$\alpha: f^*\Omega_{Y/k} \longrightarrow \Omega_{X/k}$$

is a map between locally free sheaves of the same rank and is an isomorphism at the generic point of  $X$ . Hence  $\alpha$  is injective, and taking determinants gives

$$\det(\alpha): f^*\omega_{Y/k} \longrightarrow \omega_{X/k}.$$

Equivalently,  $\det(\alpha)$  is a nonzero section of

$$\omega_{X/k} \otimes f^*\omega_{Y/k}^{-1}.$$

**Definition 5.15.** The *ramification divisor* of  $f$  is

$$R(f) = Z(\det \alpha).$$

**Proposition 5.16.** *Away from  $R(f)$ , the morphism  $f$  is unramified. Moreover,*

$$\omega_X \cong f^*\omega_Y \otimes \mathcal{O}_X(R(f)).$$

**Example 5.17** (Power maps). For

$$f: \mathbb{A}_k^1 \longrightarrow \mathbb{A}_k^1, \quad x \longmapsto x^n,$$

with  $\text{char } k \nmid n$ ,

$$f^*(dt) \longmapsto nx^{n-1} dx.$$

Thus the ramification divisor is  $(n-1)\{0\}$ .

### 5.4 The Hurwitz Formula

Let  $f: X \rightarrow Y$  be a nonconstant morphism of smooth projective integral curves, and assume  $k(X)/k(Y)$  is separable. Such a morphism is finite and surjective.

**Theorem 5.18** (Riemann–Hurwitz).

$$2g(X) - 2 = \deg(f)(2g(Y) - 2) + \deg R(f).$$

*Proof.* Take degrees in

$$\omega_X \cong f^*\omega_Y \otimes \mathcal{O}_X(R(f))$$

and use

$$\deg \omega_X = 2g(X) - 2, \quad \deg f^* \omega_Y = \deg(f) \deg \omega_Y.$$

□

**Corollary 5.19** (Lüroth's theorem). *Let  $k = \bar{k}$  and let*

$$k \subsetneq K \subset k(t)$$

*be a field of transcendence degree one over  $k$ . Then*

$$K \cong k(u).$$

*Proof.* Let  $X$  be the unique smooth projective curve with function field  $K$ . The inclusion  $K \subset k(t)$  gives a finite map

$$\mathbb{P}^1 \longrightarrow X.$$

Riemann–Hurwitz gives

$$-2 = \deg(f)(2g(X) - 2) + \deg R(f).$$

Since the ramification term is nonnegative,  $g(X) = 0$ . Over an algebraically closed field,  $X \cong \mathbb{P}^1$ , and hence  $K \cong k(u)$ . □

**Remark 5.20** (The higher-dimensional Lüroth problem). Over an algebraically closed field of characteristic zero, unirational surfaces are rational, but the analogous statement fails in dimension at least three. A classical example is a smooth cubic threefold  $X \subset \mathbb{P}_{\mathbb{C}}^4$ : it is unirational but not rational.

**Example 5.21.** There is no nontrivial connected finite étale cover of  $\mathbb{P}^1$ . Indeed, if  $f: X \rightarrow \mathbb{P}^1$  is étale, then

$$2g(X) - 2 = -2 \deg(f),$$

which forces  $\deg(f) = 1$ .

**Example 5.22** (Double covers). A degree-two map  $\mathbb{P}^1 \rightarrow \mathbb{P}^1$  has ramification divisor of degree two and, after changing coordinates, is  $z \mapsto z^2$ . A double cover of  $\mathbb{P}^1$  branched at four distinct points has genus one; a double cover branched at six distinct points has genus two.

### 5.4.1 Ramification Indices

Let  $x \in X$ ,  $y = f(x)$ , and choose uniformizers  $s$  and  $t$  in the discrete valuation rings  $\mathcal{O}_{X,x}$  and  $\mathcal{O}_{Y,y}$ . Then

$$t = us^{e_x}$$

for a unit  $u$ , and  $e_x$  is the *ramification index*.

**Definition 5.23.** The map is *ramified at  $x$*  if  $e_x > 1$ . It is *tamely ramified at  $x$*  if the residue extension is separable and  $\text{char } k(x) \nmid e_x$ .

Put

$$d_x = \text{length}_{\mathcal{O}_{X,x}}(\Omega_{X/Y,x}).$$



Then  $d_x$  is the coefficient of  $x$  in  $R(f)$ . In the tame case,

$$d_x = e_x - 1.$$

Indeed,

$$dt = d(us^{e_x}) = s^{e_x-1}(s du + e_x u ds),$$

and the factor in parentheses is a unit multiple of  $ds$  when  $e_x$  is invertible. In the wild case,  $d_x \geq e_x$  may occur.

For a tame cover over an algebraically closed field,

$$R(f) = \sum_x (e_x - 1)x,$$

and Riemann–Hurwitz becomes

$$2g(X) - 2 = \deg(f)(2g(Y) - 2) + \sum_x (e_x - 1).$$

## 5.5 Automorphisms of Curves

**Theorem 5.24** (Hurwitz bound). *Let  $X$  be a smooth projective curve over an algebraically closed field of characteristic zero, with  $g(X) \geq 2$ . Then*

$$\text{Aut}_k(X) \text{ is finite,} \quad |\text{Aut}_k(X)| \leq 84(g(X) - 1).$$

*Proof sketch.* Let  $G \subseteq \text{Aut}_k(X)$  be finite, put  $Y = X/G$ , and let  $F: X \rightarrow Y$  be the quotient map. Then

$$k(Y) = k(X)^G, \quad \deg F = |G|.$$

For a branch point  $y$ , the group  $G$  acts transitively on the fiber, all ramification indices are equal to some  $e_y$ , and

$$2g(X) - 2 = |G| \left( 2g(Y) - 2 + \sum_y \left( 1 - \frac{1}{e_y} \right) \right).$$

If  $g(Y) \geq 2$ , the expression in parentheses is at least 2. If  $g(Y) = 1$ , it is at least  $1/2$ . If  $g(Y) = 0$ , positivity and an elementary analysis of the possible ramification indices show that it is at least  $1/42$ , with equality for the signature  $(2, 3, 7)$ . Thus  $|G| \leq 84(g(X) - 1)$ . A separate standard argument shows that  $\text{Aut}_k(X)$  itself is finite, so take  $G = \text{Aut}_k(X)$ .  $\square$

**Example 5.25** (The Klein quartic). The curve

$$x^3y + y^3z + z^3x = 0 \subset \mathbb{P}^2$$

has genus 3 and automorphism group of order  $168 = 84(3-1)$ . Curves attaining the Hurwitz bound are called *Hurwitz curves*.

**Remark 5.26** (Belyi's theorem). A smooth projective complex curve is defined over  $\overline{\mathbb{Q}}$  if and only if it admits a finite map to  $\mathbb{P}^1$  branched over at most three points. In particular, Hurwitz

curves are defined over  $\overline{\mathbb{Q}}$ .

## 5.6 Local Structure and Infinitesimal Lifting

**Theorem 5.27** (Smooth morphisms are étale-locally projections). *If  $f: X \rightarrow Y$  is smooth of relative dimension  $d$ , then locally on  $X$  there is a factorization*

$$X \longrightarrow \mathbb{A}_Y^d \longrightarrow Y$$

*whose first arrow is étale and whose second arrow is the projection.*

**Definition 5.28** (Standard étale algebra). Let  $A$  be a ring and let  $F \in A[t]$  be monic. The  $A$ -algebra

$$B = (A[t]/(F))_{F'}$$

is *standard étale*; geometrically,

$$\mathrm{Spec} B = V(F) \setminus V(F') \longrightarrow \mathrm{Spec} A$$

is étale.

Every étale morphism is locally modeled by a standard étale morphism. The passage from a general quasi-finite morphism to this form is closely related to Zariski's Main Theorem.

### 5.6.1 Formal Smoothness

Let  $Z_0 \hookrightarrow Z$  be a closed immersion defined by a nilpotent ideal.

**Definition 5.29.** A morphism  $f: X \rightarrow Y$  is:

- *formally smooth* if every diagram

$$\begin{array}{ccc} Z_0 & \longrightarrow & X \\ \downarrow & \nearrow \exists & \downarrow f \\ Z & \longrightarrow & Y \end{array}$$

admits a lift;

- *formally unramified* if such a lift is unique whenever it exists;
- *formally étale* if such a lift exists uniquely.

**Theorem 5.30.** *Under the appropriate finite-presentation hypotheses:*

$$\begin{aligned} f \text{ is smooth} &\iff f \text{ is formally smooth,} \\ f \text{ is unramified} &\iff f \text{ is formally unramified,} \\ f \text{ is étale} &\iff f \text{ is formally étale.} \end{aligned}$$

For  $Z = \operatorname{Spec}(k[\varepsilon]/(\varepsilon^2))$ , the lifting problem is the tangent-space lifting problem. This explains the tangent-space characterizations of smooth, unramified, and étale morphisms between smooth schemes.

## 5.7 The Small Étale Site

**Definition 5.31.** The *small étale site*  $X_{\text{ét}}$  has:

- objects: étale morphisms  $U \rightarrow X$ ;
- morphisms:  $X$ -morphisms  $U \rightarrow V$ ;
- coverings: families  $\{U_i \rightarrow U\}$  of étale morphisms that are jointly surjective.

Open immersions are étale, so the Zariski site embeds in the étale site. Fiber products exist in  $X_{\text{ét}}$  because étale morphisms are stable under base change.

**Definition 5.32.** An *étale presheaf* is a contravariant functor

$$\mathcal{F}: X_{\text{ét}}^{\text{op}} \longrightarrow \mathbf{Ab}.$$

It is a sheaf if, for every étale covering  $\{U_i \rightarrow U\}$ , the sequence

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \times_U U_j)$$

is exact.

The category  $\operatorname{Sh}(X_{\text{ét}})$  is abelian and has enough injectives. Thus one defines

$$H_{\text{ét}}^i(X, \mathcal{F}) = R^i \Gamma(X_{\text{ét}}, \mathcal{F}).$$

Epimorphisms of étale sheaves need only be locally surjective in the étale topology; they need not be surjective on sections over every object.

### 5.7.1 Geometric Stalks

Let  $\bar{x} \rightarrow X$  be a geometric point. The stalk is

$$\mathcal{F}_{\bar{x}} = \varinjlim_{(U, \bar{u})} \mathcal{F}(U),$$

where the limit ranges over pointed étale neighborhoods of  $\bar{x}$ . For the structure sheaf,

$$\mathcal{O}_{X_{\text{ét}}, \bar{x}} \cong \mathcal{O}_{X, \bar{x}}^{\text{sh}},$$

the strict henselization. In particular, for  $X = \operatorname{Spec} k$ , this stalk is a separable closure  $k^{\text{sep}}$ .

## 5.8 Examples of Étale Sheaves

The étale site is subcanonical: every  $X$ -scheme  $G$  defines a sheaf

$$U \longmapsto \mathrm{Hom}_X(U, G).$$

In particular,

$$\mathbb{G}_m(U) = \Gamma(U, \mathcal{O}_U)^\times, \quad \mathbb{G}_a(U) = \Gamma(U, \mathcal{O}_U),$$

and

$$\mu_n(U) = \{a \in \Gamma(U, \mathcal{O}_U)^\times : a^n = 1\}.$$

**Theorem 5.33** (Kummer sequence). *If  $n$  is invertible on  $X$ , then*

$$1 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \xrightarrow{(\cdot)^n} \mathbb{G}_m \longrightarrow 1$$

*is exact on  $X_{\text{ét}}$ .*

Surjectivity on the right is étale-local: an invertible function gains an  $n$ -th root after the finite étale extension defined by  $T^n - a$ .

**Proposition 5.34.**

$$H_{\text{ét}}^1(X, \mathbb{G}_m) \cong \mathrm{Pic}(X).$$

The Kummer sequence gives

$$0 \longrightarrow \Gamma(X, \mathcal{O}_X^\times) / \Gamma(X, \mathcal{O}_X^\times)^n \longrightarrow H_{\text{ét}}^1(X, \mu_n) \longrightarrow \mathrm{Pic}(X)[n] \longrightarrow 0.$$

Equivalently, an element of  $H_{\text{ét}}^1(X, \mu_n)$  may be represented by a pair

$$(\mathcal{L}, \varphi), \quad \varphi: \mathcal{L}^{\otimes n} \xrightarrow{\sim} \mathcal{O}_X.$$

Nonabelian cohomology gives

$$H_{\text{ét}}^1(X, GL_n) \cong \{\text{rank-}n \text{ vector bundles on } X\} / \cong.$$

Étale-local freeness of a coherent module descends to ordinary local freeness. In contrast,

$$H_{\text{ét}}^1(X, PGL_n)$$

classifies Brauer–Severi schemes, which may be étale-locally but not Zariski-locally isomorphic to  $\mathbb{P}^{n-1}$ .

**Example 5.35** (An étale torsor that is not Zariski-trivial). Let  $k$  have characteristic different from 2. The map

$$\mathbb{G}_{m,k} \longrightarrow \mathbb{G}_{m,k}, \quad z \longmapsto z^2$$

is a finite étale  $\mu_2$ -torsor. It is not trivial over any nonempty Zariski open subset because  $t$  is not a square in  $k(t)$ , but it becomes trivial after pulling back along itself.

**Example 5.36** (An isogeny of elliptic curves). Let  $E$  be an elliptic curve and let  $P \in E[2]$  be

nonzero, with  $\text{char } k \neq 2$ . Translation by  $P$  defines a free action of  $\mathbb{Z}/2\mathbb{Z}$ , and

$$E \longrightarrow E/\langle P \rangle$$

is a finite étale cover of degree two. It is étale-locally trivial but not Zariski-locally a disjoint union of two copies of the base.

## 5.9 Comparison over the Complex Numbers

**Theorem 5.37** (Artin comparison). *Let  $X$  be of finite type over  $\mathbb{C}$ , and let  $\Lambda$  be a finite abelian group. Then*

$$H_{\text{ét}}^i(X, \Lambda) \cong H^i(X^{\text{an}}, \Lambda).$$

Finite étale covers of  $X$  analytify to finite topological covering spaces of  $X^{\text{an}}$ . For coefficients such as  $\mathbb{G}_m$ , étale and analytic cohomology need not agree in general; GAGA supplies comparison results when  $X$  is proper.

## 5.10 The Étale Fundamental Group

Let  $X$  be connected and choose a geometric point  $\bar{x} \rightarrow X$ . Let  $\text{FÉt}(X)$  be the category of finite étale covers of  $X$ . The fiber functor is

$$F_{\bar{x}}: \text{FÉt}(X) \longrightarrow \text{FinSets}, \quad (Y \rightarrow X) \longmapsto \text{Hom}_X(\bar{x}, Y).$$

**Definition 5.38.** The *étale fundamental group* is the profinite group

$$\pi_1^{\text{ét}}(X, \bar{x}) = \text{Aut}(F_{\bar{x}}).$$

There is an equivalence

$$\text{FÉt}(X) \simeq \{\text{finite continuous } \pi_1^{\text{ét}}(X, \bar{x})\text{-sets}\}.$$

The inverse system playing the role of a universal cover is generally not represented by a scheme, which is why the fundamental group is profinite.

**Example 5.39.** For  $X = \mathbb{G}_{m, \mathbb{C}}$ , the maps

$$\mathbb{G}_m \longrightarrow \mathbb{G}_m, \quad z \longmapsto z^n,$$

are finite étale covers. They correspond analytically to the standard finite coverings of the circle.

**Theorem 5.40** (Riemann existence). *If  $X$  is a connected complex variety, then*

$$\pi_1^{\text{ét}}(X, \bar{x}) \cong \widehat{\pi_1^{\text{top}}(X^{\text{an}}, x)},$$

*the profinite completion of the topological fundamental group.*

The profinite completion can forget information. Serre constructed conjugate smooth projective complex varieties whose étale fundamental groups are isomorphic, while their topological fundamental groups are not isomorphic.

## Chapter 6

# Blow-ups, Varieties, and Function Fields

### 6.1 The universal property of the blow-up

Let  $X$  be a scheme and let  $Z \subseteq X$  be a closed subscheme defined by a quasi-coherent ideal sheaf  $\mathcal{I}$  of finite type.

**Definition 6.1.** The *blow-up of  $X$  along  $Z$*  is a morphism

$$\beta: \mathrm{Bl}_Z(X) \longrightarrow X$$

with the following properties:

1. the ideal  $\mathcal{I}\mathcal{O}_{\mathrm{Bl}_Z(X)}$  is invertible, so its zero scheme is an effective Cartier divisor;
2. for every morphism  $f: W \rightarrow X$  such that  $\mathcal{I}\mathcal{O}_W$  is invertible, there is a unique morphism  $g: W \rightarrow \mathrm{Bl}_Z(X)$  satisfying  $\beta \circ g = f$ .

Thus the blow-up is terminal among  $X$ -schemes on which the ideal of the center becomes an invertible ideal. The effective Cartier divisor

$$E = \beta^{-1}(Z)$$

is called the *exceptional divisor*. The universal property gives a Cartesian-looking diagram

$$\begin{array}{ccc} E & \hookrightarrow & \mathrm{Bl}_Z(X) \\ \downarrow & & \downarrow \beta \\ Z & \hookrightarrow & X, \end{array}$$

where  $E$  is the scheme-theoretic inverse image of  $Z$ .

*Remark 6.2.* For a morphism  $f: W \rightarrow X$ , the notation  $\mathcal{I}\mathcal{O}_W$  means the image of

$$f^*\mathcal{I} \longrightarrow \mathcal{O}_W.$$

The pullback  $f^*\mathcal{I}$  need not inject into  $\mathcal{O}_W$ . It is the image ideal, rather than the abstract pullback module, that defines  $f^{-1}(Z)$ .

**Definition 6.3.** Let  $Y \subseteq X$  be a closed subscheme.

- The *total transform* of  $Y$  is the scheme-theoretic inverse image  $\beta^{-1}(Y)$ .
- If  $Y \not\subseteq Z$ , the *strict transform* of  $Y$  is the closure in  $\text{Bl}_Z(X)$  of  $\beta^{-1}(Y \setminus Z)$ .

**Proposition 6.4** (Immediate consequences of the universal property). *Let  $\beta: \text{Bl}_Z(X) \rightarrow X$  be a blow-up.*

1. *If  $Z$  is already an effective Cartier divisor on  $X$ , then  $\beta$  is an isomorphism.*
2. *Blow-ups are local on the base: for every open  $U \subseteq X$ ,*

$$\beta^{-1}(U) \simeq \text{Bl}_{Z \cap U}(U).$$

3. *The restriction*

$$\text{Bl}_Z(X) \setminus E \longrightarrow X \setminus Z$$

*is an isomorphism.*

*Proof.* If  $\mathcal{I}$  is already invertible, the identity map of  $X$  satisfies the defining property and is therefore inverse to the blow-up map. The other assertions follow by applying the same universal property after restriction to an open subscheme.  $\square$

## 6.2 Construction from the Rees algebra

The universal object exists and has a concrete description.

**Theorem 6.5.** *Let  $\mathcal{I} \subseteq \mathcal{O}_X$  be a quasi-coherent ideal of finite type. Then*

$$\text{Bl}_Z(X) = \text{Proj}_X \left( \bigoplus_{n \geq 0} \mathcal{I}^n \right),$$

*where  $Z = V(\mathcal{I})$ . The graded algebra*

$$\mathcal{R}(\mathcal{I}) = \bigoplus_{n \geq 0} \mathcal{I}^n$$

*is called the Rees algebra of  $\mathcal{I}$ .*

Under the convention used here, the tautological invertible sheaf satisfies

$$\mathcal{I}\mathcal{O}_{\text{Bl}_Z(X)} \simeq \mathcal{O}_{\text{Bl}_Z(X)}(1) \simeq \mathcal{O}_{\text{Bl}_Z(X)}(-E).$$

Consequently,

$$\mathcal{O}_{\text{Bl}_Z(X)}(E) \simeq \mathcal{O}_{\text{Bl}_Z(X)}(-1).$$

*Idea of the construction.* The assertion is local on  $X$ . Work over an affine open  $X = \text{Spec } A$ , and write  $I = (a_0, \dots, a_m)$ . The surjection

$$A[T_0, \dots, T_m] \longrightarrow \mathcal{R}(I), \quad T_i \longmapsto a_i$$



is a homomorphism of graded  $A$ -algebras. It realizes the blow-up as a closed subscheme

$$\mathrm{Bl}_I(\mathrm{Spec} A) \hookrightarrow \mathbb{P}_A^m.$$

On the standard chart  $D_+(a_i)$ , the ideal  $I$  becomes principal, generated by  $a_i$ . Hence the pullback of  $I$  is invertible.

Conversely, if  $f: W \rightarrow X$  makes  $f^*I \rightarrow \mathcal{O}_W$  invertible, the quotient  $f^*I \rightarrow \mathcal{O}_W$  determines a unique morphism from  $W$  to the relative Proj. This proves the universal property.  $\square$

**Corollary 6.6.** *The blow-up morphism is projective. If  $X$  is projective over a base  $S$ , then  $\mathrm{Bl}_Z(X)$  is projective over  $S$ .*

**Proposition 6.7.** *If  $X$  is reduced, then  $\mathrm{Bl}_Z(X)$  is reduced. If  $X$  is integral, then  $\mathrm{Bl}_Z(X)$  is integral.*

*Proof.* On an affine open  $X = \mathrm{Spec} A$ , the Rees algebra is a graded subalgebra of  $A[t]$ . If  $A$  is reduced, respectively a domain, then so is the Rees algebra. The corresponding statement passes to its Proj.  $\square$

## 6.3 The blow-up of affine space

### 6.3.1 The origin in the affine plane

Let  $X = \mathbb{A}_k^2 = \mathrm{Spec} k[x, y]$  and let  $Z = V(x, y)$ . The blow-up is the closed subscheme

$$\mathrm{Bl}_0(\mathbb{A}^2) = \{((x, y), [u : v]) \in \mathbb{A}^2 \times \mathbb{P}^1 \mid xv - yu = 0\}.$$

It is covered by two affine charts:

| chart      | coordinate | equation  |
|------------|------------|-----------|
| $u \neq 0$ | $w = v/u$  | $y = xw,$ |
| $v \neq 0$ | $z = u/v$  | $x = yz.$ |

The exceptional divisor is

$$E = \beta^{-1}(0) \simeq \mathbb{P}^1.$$

Its points record tangent directions through the origin.

More generally, the blow-up of the origin in  $\mathbb{A}_k^n$  is the closed subscheme of  $\mathbb{A}^n \times \mathbb{P}^{n-1}$  defined by

$$x_i u_j - x_j u_i = 0 \quad (1 \leq i, j \leq n).$$

The exceptional divisor is  $\mathbb{P}^{n-1}$ .

### 6.3.2 Total and strict transforms

Consider the line  $L_t = V(y - tx)$  in  $\mathbb{A}^2$ . On the chart  $u \neq 0$ , where  $y = xw$ , its total transform is

$$x(w - t) = 0.$$

Thus it is the union of the exceptional divisor  $V(x)$  and the strict transform  $V(w - t)$ . The strict transform meets  $E$  at the point representing the slope  $t$ .

**Proposition 6.8.** *Let  $Y \subseteq X$  be a closed subscheme not contained in  $Z$ . Its strict transform in  $\mathrm{Bl}_Z(X)$  is canonically isomorphic to*

$$\mathrm{Bl}_{Y \cap Z}(Y),$$

*where the center is defined by the image of  $\mathcal{I}_Z \mathcal{O}_Y \rightarrow \mathcal{O}_Y$ .*

**Remark 6.9.** The total transform remembers every exceptional component created by the pullback of the defining equations. The strict transform removes those exceptional factors and retains the closure of the part lying over  $X \setminus Z$ .

## 6.4 Geometry of blow-ups

If  $X$  is integral and  $Z$  is nowhere dense, then  $\beta: \mathrm{Bl}_Z(X) \rightarrow X$  is a projective birational morphism: it is an isomorphism over the dense open set  $X \setminus Z$ . It is also surjective. Blow-ups are generally not flat, since the fibers can jump in dimension; for example, the blow-up of a point on a surface has one-dimensional exceptional fiber and zero-dimensional generic fiber.

### 6.4.1 Resolving plane curve singularities

Let

$$C = V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2.$$

The origin is a node. On the blow-up chart  $y = xw$ , the total transform has equation

$$x^2(w^2 - x - 1) = 0.$$

After removing the exceptional factor  $x^2$ , the strict transform is

$$w^2 = x + 1.$$

It is smooth near the exceptional divisor and meets  $E = V(x)$  in the two points  $w = \pm 1$ , corresponding to the two tangent directions of the node.

For a cusp such as  $y^2 = x^3$ , one blow-up replaces the cusp by a strict transform with lower-order singular behavior. Repeating the process eventually resolves any plane curve singularity. Equations such as  $y^3 = x^5$  illustrate why more than one blow-up may be necessary.

### 6.4.2 Smooth centers

**Theorem 6.10.** *Let  $X$  be smooth over a field  $k$ , and let  $Z \subseteq X$  be a smooth closed subscheme. Then  $\mathrm{Bl}_Z(X)$  is smooth over  $k$ . Moreover,*

$$E \simeq \mathbb{P}_Z(N_{Z/X}) \simeq \mathrm{Proj}_Z(\mathrm{Sym}_{\mathcal{O}_Z}(\mathcal{I}/\mathcal{I}^2)),$$

*with the projectivization convention determined by one-dimensional quotients, and*

$$N_{E/\mathrm{Bl}_Z(X)} \simeq \mathcal{O}_E(-1).$$

*Sketch.* Smoothness is local in the étale topology. Near a point of  $Z$ , one may choose regular parameters such that  $Z$  is cut out by part of a regular system of parameters. The standard affine charts of the blow-up are then affine spaces with regular coordinates, hence smooth. Since  $Z \hookrightarrow X$  is a regular immersion,

$$\mathrm{gr}_{\mathcal{I}}(\mathcal{O}_X) \simeq \mathrm{Sym}_{\mathcal{O}_Z}(\mathcal{I}/\mathcal{I}^2).$$

Taking relative Proj identifies the exceptional divisor with the projectivized normal bundle. The ideal of  $E$  restricts to the tautological line bundle  $\mathcal{O}_E(1)$ , so its normal bundle is  $\mathcal{O}_E(-1)$ .  $\square$

### 6.4.3 Resolving projection from a point

Fix  $p \in \mathbb{P}^2$ . Projection from  $p$  defines a rational map

$$\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$$

that sends a point  $q \neq p$  to the line through  $p$  and  $q$ . Blowing up  $p$  resolves the indeterminacy:

$$\begin{array}{ccc} \mathrm{Bl}_p(\mathbb{P}^2) & \longrightarrow & \mathbb{P}^1 \\ \beta \downarrow & \nearrow & \\ \mathbb{P}^2 & & \end{array}$$

The exceptional curve  $E \simeq \mathbb{P}^1$  parametrizes the lines through  $p$ . In fact,  $\mathrm{Bl}_p(\mathbb{P}^2)$  is the Hirzebruch surface  $\mathbb{F}_1$ , and the morphism to  $\mathbb{P}^1$  has as fibers the strict transforms of those lines.

## 6.5 Classical varieties and schemes

Let  $k$  be an algebraically closed field. A classical affine variety is an irreducible algebraic subset of affine space, equipped with its sheaf of regular functions. Projective and quasi-projective varieties are defined similarly. A regular function is locally a quotient of polynomial functions whose denominator does not vanish, and morphisms are maps locally given by regular functions.

There is a fully faithful passage from classical varieties to schemes. For a classical variety  $V$ , define  $T(V)$  as follows:

- its points are the irreducible closed subsets of  $V$ ;
- the topology is generated so that a closed subset of  $V$  gives the set of irreducible closed subsets contained in it;
- the map

$$V \longrightarrow T(V), \quad x \longmapsto \overline{\{x\}},$$

identifies the classical points with the closed points of  $T(V)$ ;

- the structure sheaf is the unique one whose restriction along this dense inclusion recovers the classical sheaf of regular functions.

If  $V$  is affine with coordinate ring  $A$ , then

$$T(V) \simeq \operatorname{Spec} A.$$

Indeed, irreducible closed subsets correspond to prime ideals, whereas classical points correspond to maximal ideals. Thus passing from  $\operatorname{MaxSpec} A$  to  $\operatorname{Spec} A$  adds generic points.

**Definition 6.11.** An *abstract variety over  $k$*  is an integral, separated scheme of finite type over  $k$ .

The construction  $T$  gives a fully faithful functor from classical quasi-projective varieties to abstract varieties. It is not essentially surjective in general: not every abstract variety is quasi-projective. For projective varieties, however, the classical and scheme-theoretic categories agree.

Over  $k = \mathbb{C}$ , a projective variety also has an associated compact complex analytic space. Serre's GAGA theorem compares algebraic and analytic coherent sheaves and identifies algebraic morphisms with holomorphic ones in the proper setting.

## 6.6 Function fields and rational maps

Let  $X$  be an integral variety. Its *function field* is

$$K(X) = \mathcal{O}_{X,\eta},$$

where  $\eta$  is the generic point. Equivalently, if  $\operatorname{Spec} A \subseteq X$  is any nonempty affine open subset, then

$$K(X) = \operatorname{Frac}(A).$$

**Definition 6.12.** A *rational map*  $f: X \dashrightarrow Y$  is an equivalence class of morphisms  $U \rightarrow Y$ , where  $U \subseteq X$  is a nonempty open subset. Two representatives are equivalent if they agree on a nonempty open subset.

A rational map is *dominant* if its image is dense. A dominant rational map

$$f: X \dashrightarrow Y$$

induces an injective homomorphism of fields

$$f^*: K(Y) \hookrightarrow K(X).$$

Conversely, every  $k$ -embedding  $K(Y) \hookrightarrow K(X)$  comes from a unique dominant rational map. Hence dominant rational maps are contravariantly equivalent to inclusions of finitely generated field extensions of  $k$ .

**Theorem 6.13.** For integral varieties  $X$  and  $Y$ , the following are equivalent:

1.  $X$  and  $Y$  are birational;
2. there exist nonempty open subsets  $U \subseteq X$  and  $V \subseteq Y$  with  $U \simeq V$ ;

3. *their function fields are isomorphic over  $k$ :*

$$K(X) \simeq K(Y).$$

If  $X \dashrightarrow Y$  is dominant and  $Y \dashrightarrow Z$  is rational, their composition is defined on a nonempty open subset. This dominance hypothesis ensures that the generic point of  $X$  lands in the domain of definition of the second map.

**Proposition 6.14.** *Every  $n$ -dimensional variety over an algebraically closed field is birational to a hypersurface in  $\mathbb{A}^{n+1}$ , and hence also to a projective hypersurface after taking a closure.*

*Idea.* The extension  $K(X)/k$  has transcendence degree  $n$ . Choose a transcendence basis  $t_1, \dots, t_n$ . Since  $k$  is perfect, the remaining finite extension is generated by one element  $a$ . The minimal polynomial of  $a$  over  $k(t_1, \dots, t_n)$ , after clearing denominators, gives one equation in  $n + 1$  variables with function field  $K(X)$ .  $\square$

**Remark 6.15** (The Lüroth problem). It is natural to ask whether every unirational variety is rational. For curves the answer is yes by Lüroth's theorem, and for surfaces over an algebraically closed field of characteristic zero it follows from Castelnuovo's criterion. In higher dimensions the answer is no: there exist unirational varieties whose function fields are not purely transcendental.

## Chapter 7

# Completion and Formal Functions

### 7.1 Filtered modules and adic completion

Let  $A$  be a ring and let  $M$  be an  $A$ -module equipped with a descending filtration

$$M = M_0 \supseteq M_1 \supseteq M_2 \supseteq \cdots.$$

**Definition 7.1.** The *associated graded module* and the *completion* of  $M$  are

$$\mathrm{gr}(M) = \bigoplus_{i \geq 0} M_i / M_{i+1}, \quad \widehat{M} = \varprojlim_i M / M_i.$$

The filtration defines a topology on  $M$ , with the cosets  $x + M_i$  as a basis of open neighborhoods. It is Hausdorff precisely when

$$\bigcap_{i \geq 0} M_i = 0.$$

Indeed, two distinct elements can be separated exactly when their difference does not belong to every  $M_i$ .

**Definition 7.2.** Two filtrations  $\{M_i\}$  and  $\{M'_i\}$  are *equivalent* if they are cofinal: for every  $i$  there is a  $j$  such that  $M'_j \subseteq M_i$ , and conversely.

Equivalent filtrations define the same topology and canonically isomorphic completions. The main case is the  $I$ -adic filtration

$$M_i = I^i M$$

for an ideal  $I \subseteq A$ . We then write

$$\widehat{M} = \varprojlim_n M / I^n M, \quad \widehat{A} = \varprojlim_n A / I^n.$$

The module  $\widehat{M}$  is naturally an  $\widehat{A}$ -module, and the canonical map  $M \rightarrow \widehat{M}$  is injective if and only if the  $I$ -adic topology on  $M$  is Hausdorff.

**Example 7.3.**

1. The  $(p)$ -adic completion of  $\mathbb{Z}$  is the ring of  $p$ -adic integers:

$$\widehat{\mathbb{Z}} \simeq \mathbb{Z}_p.$$

2. If  $A = k[x_1, \dots, x_n]$  and  $\mathfrak{m} = (x_1, \dots, x_n)$ , then

$$\widehat{A} \simeq k[[x_1, \dots, x_n]].$$

If  $\mathfrak{m}$  is a maximal ideal of a Noetherian ring  $A$ , then completion at  $\mathfrak{m}$  commutes with localization:

$$\widehat{A}^{\mathfrak{m}} \simeq \widehat{A_{\mathfrak{m}}}^{\mathfrak{m}A_{\mathfrak{m}}}.$$

## 7.2 The Artin–Rees lemma

**Theorem 7.4** (Artin–Rees). *Let  $A$  be Noetherian,  $I \subseteq A$  an ideal,  $M$  a finite  $A$ -module, and  $N \subseteq M$  a submodule. There is an integer  $c \geq 0$  such that*

$$I^n M \cap N = I^{n-c}(I^c M \cap N) \quad (n \geq c).$$

*In particular, the filtrations*

$$\{I^n N\}_{n \geq 0} \quad \text{and} \quad \{I^n M \cap N\}_{n \geq 0}$$

*on  $N$  are equivalent.*

*Proof.* Consider the Rees algebra and the corresponding graded modules

$$S = \bigoplus_{n \geq 0} I^n, \quad \widetilde{M} = \bigoplus_{n \geq 0} I^n M, \quad \widetilde{N} = \bigoplus_{n \geq 0} (I^n M \cap N).$$

If  $I = (a_1, \dots, a_r)$ , then  $S$  is a quotient of  $A[T_1, \dots, T_r]$ , and is therefore Noetherian. The module  $\widetilde{M}$  is finite over  $S$ , so its graded submodule  $\widetilde{N}$  is finite as well. Choose homogeneous generators of degrees at most  $c$ . Comparing the degree- $n$  components for  $n \geq c$  gives

$$I^n M \cap N = I^{n-c}(I^c M \cap N).$$

□

**Theorem 7.5** (Krull intersection theorem). *If  $(A, \mathfrak{m})$  is a Noetherian local ring and  $M$  is a finite  $A$ -module, then*

$$\bigcap_{n \geq 0} \mathfrak{m}^n M = 0.$$

*In particular,  $A \rightarrow \widehat{A}$  is injective.*

*Proof.* Put  $N = \bigcap_n \mathfrak{m}^n M$ . Artin–Rees implies, for all sufficiently large  $n$ ,

$$N = \mathfrak{m}^n M \cap N = \mathfrak{m}(\mathfrak{m}^{n-1} M \cap N) = \mathfrak{m}N.$$

Since  $N$  is finite, Nakayama’s lemma gives  $N = 0$ .

□

### 7.3 Fundamental properties of completion

**Theorem 7.6.** *Let  $A$  be Noetherian, let  $I \subseteq A$ , and let completion be taken with respect to  $I$ .*

1. *Completion is exact on finite  $A$ -modules.*
2. *For every finite  $A$ -module  $M$ , the natural map*

$$M \otimes_A \hat{A} \xrightarrow{\sim} \widehat{M}$$

*is an isomorphism.*

3. *The homomorphism  $A \rightarrow \hat{A}$  is flat.*
4. *The ring  $\hat{A}$  is Noetherian.*
5. *If  $(A, \mathfrak{m})$  is local and  $I = \mathfrak{m}$ , then  $\hat{A}$  is local, with maximal ideal  $\hat{\mathfrak{m}} = \mathfrak{m}\hat{A}$ , and*

$$A/\mathfrak{m}^n \simeq \hat{A}/\hat{\mathfrak{m}}^n, \quad \text{gr}_{\mathfrak{m}}(A) \simeq \text{gr}_{\hat{\mathfrak{m}}}(\hat{A}).$$

*Proof sketch.* For an exact sequence

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0,$$

the induced filtration on  $M_1$  is  $\{I^n M_2 \cap M_1\}$ . Artin–Rees says that this filtration is equivalent to the  $I$ -adic filtration. The quotient transition maps are surjective, so taking inverse limits preserves exactness here.

A finite presentation of  $M$ , together with exactness, identifies  $\widehat{M}$  with  $M \otimes_A \hat{A}$ . Exactness of tensoring finite modules then gives flatness; over a Noetherian ring this criterion is sufficient. The Noetherian property of  $\hat{A}$  follows from the associated graded ring, or from a power-series presentation after choosing generators of  $I$ . The final statements follow by completing  $0 \rightarrow \mathfrak{m}^n \rightarrow A \rightarrow A/\mathfrak{m}^n \rightarrow 0$ .  $\square$

**Corollary 7.7.** *For a Noetherian local ring  $(A, \mathfrak{m})$ , the following are equivalent:*

1.  *$A$  is regular;*
2.  *$\hat{A}$  is regular;*
3. *if  $k = A/\mathfrak{m}$  and  $d = \dim A$ , then*

$$\text{gr}_{\mathfrak{m}}(A) \simeq k[x_1, \dots, x_d].$$

*Proof.* Completion preserves the dimension and the embedding dimension  $\dim_k \mathfrak{m}/\mathfrak{m}^2$ . A local ring is regular exactly when these two numbers agree. Equivalently, a regular system of parameters induces isomorphisms

$$\text{Sym}_k^n(\mathfrak{m}/\mathfrak{m}^2) \xrightarrow{\sim} \mathfrak{m}^n/\mathfrak{m}^{n+1}$$

for every  $n$ .  $\square$

**Theorem 7.8** (Equicharacteristic Cohen structure theorem). *Let  $(A, \mathfrak{m}, k)$  be a complete Noethe-*



rian local ring containing a coefficient field  $k \subseteq A$  that maps isomorphically onto  $A/\mathfrak{m}$ . Then

$$A \simeq k[[x_1, \dots, x_r]]/J$$

for some ideal  $J$ , where  $r = \dim_k \mathfrak{m}/\mathfrak{m}^2$ . If  $A$  is regular of dimension  $d$ , then

$$A \simeq k[[x_1, \dots, x_d]].$$

## 7.4 Faithfully flat completion

**Definition 7.9.** An  $A$ -algebra  $B$  is *faithfully flat* if a sequence of  $A$ -modules is exact if and only if it becomes exact after tensoring with  $B$ .

**Proposition 7.10.** Let  $\varphi: A \rightarrow B$  be a ring homomorphism.

1. If  $\varphi$  is faithfully flat, then it is injective.
2. If  $\varphi$  is faithfully flat and  $J \subseteq A$ , then

$$JB \cap A = J.$$

3. A flat local homomorphism of local rings is faithfully flat.
4. If  $A$  is Noetherian local, then  $A \hookrightarrow \hat{A}$  is faithfully flat.

**Example 7.11** (A node in formal coordinates). Assume  $\text{char } k \neq 2$ . Near the origin, the curve

$$y^2 = x^3 + x^2 = x^2(1+x)$$

is formally isomorphic to the ordinary node. Indeed,  $1+x$  has a square root  $u \in k[[x]]^\times$  by Hensel's lemma, and the change of coordinate  $y \mapsto y/u$  transforms the completed local ring into

$$k[[x, y]]/(y^2 - x^2) \simeq k[[s, t]]/(st).$$

The corresponding Zariski local rings need not be isomorphic; formal completion forgets information outside the infinitesimal neighborhood.

**Theorem 7.12** (Formal criterion for étaleness). Let  $\varphi: (A, \mathfrak{m}_A, k) \rightarrow (B, \mathfrak{m}_B, k)$  be a local homomorphism of Noetherian local rings, essentially of finite type, inducing the identity on the residue field. Then

$$\varphi \text{ is étale} \iff \hat{\varphi}: \hat{A} \xrightarrow{\sim} \hat{B}.$$

*Proof.* If  $\varphi$  is étale, it is flat and unramified. Thus  $\mathfrak{m}_A B = \mathfrak{m}_B$ , and faithful flatness gives

$$A \cap \mathfrak{m}_B^n = A \cap \mathfrak{m}_A^n B = \mathfrak{m}_A^n.$$

Nakayama's lemma shows that  $A/\mathfrak{m}_A^n \rightarrow B/\mathfrak{m}_B^n$  is surjective, hence it is an isomorphism. Passing to the inverse limit proves that  $\hat{\varphi}$  is an isomorphism.

Conversely, an isomorphism of completions gives

$$\mathfrak{m}_A/\mathfrak{m}_A^2 \simeq \mathfrak{m}_B/\mathfrak{m}_B^2,$$

so Nakayama's lemma yields  $\mathfrak{m}_A B = \mathfrak{m}_B$ ; this is the unramified condition under the residue-field hypothesis. Since both maps to the completions are faithfully flat, flatness descends from the isomorphism  $\widehat{A} \simeq \widehat{B}$ . Therefore  $\varphi$  is étale.  $\square$

## 7.5 The theorem on formal functions

Let  $f: X \rightarrow Y$  be proper, with  $X$  and  $Y$  locally Noetherian, and let  $\mathcal{F}$  be coherent. Fix  $y \in Y$ , put

$$A = \mathcal{O}_{Y,y}, \quad \mathfrak{m} = \mathfrak{m}_y,$$

and define the infinitesimal fibers

$$X_n = X \times_Y \operatorname{Spec}(A/\mathfrak{m}^{n+1}), \quad \mathcal{F}_n = \mathcal{F}|_{X_n}.$$

**Theorem 7.13** (The theorem on formal functions). *For every  $i \geq 0$ , the natural homomorphism is an isomorphism:*

$$(\widehat{R^i f_* \mathcal{F}})_y \xrightarrow{\sim} \varprojlim_n H^i(X_n, \mathcal{F}_n).$$

The theorem replaces the generally false expectation that cohomology commutes directly with a nonflat fiber. The entire tower of infinitesimal neighborhoods contains the missing information.

*Outline of proof.* After replacing  $Y$  by an affine neighborhood of  $y$  and then by  $\operatorname{Spec} A$ , properness gives

$$(R^i f_* \mathcal{F})^\sim \simeq \widetilde{H^i(X, \mathcal{F})}.$$

Since the closed immersion  $j_n: X_n \hookrightarrow X$  is affine,

$$H^i(X_n, \mathcal{F}_n) \simeq H^i(X, \mathcal{F}/\mathfrak{m}^{n+1} \mathcal{F}).$$

The short exact sequences

$$0 \longrightarrow \mathfrak{m}^{n+1} \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}/\mathfrak{m}^{n+1} \mathcal{F} \longrightarrow 0$$

produce comparison maps

$$H^i(X, \mathcal{F})/\mathfrak{m}^{n+1} H^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F}/\mathfrak{m}^{n+1} \mathcal{F}).$$

The key point is that the filtration on  $H^i(X, \mathcal{F})$  given by the images of

$$H^i(X, \mathfrak{m}^n \mathcal{F}) \longrightarrow H^i(X, \mathcal{F})$$

is equivalent to its  $\mathfrak{m}$ -adic filtration. This is a cohomological form of Artin–Rees, proved using the Rees algebra and the finiteness theorem for proper morphisms. The remaining kernels in the long exact cohomology sequences form a pro-zero system. Taking inverse limits therefore gives the asserted isomorphism.  $\square$

**Corollary 7.14.** *Suppose every fiber of a proper morphism  $f: X \rightarrow Y$  has dimension at most  $r$ .*

Then for every coherent sheaf  $\mathcal{F}$ ,

$$R^i f_* \mathcal{F} = 0 \quad (i > r).$$

*Proof.* The infinitesimal fibers  $X_n$  have the same underlying topological space as  $X_y$ , so their coherent cohomology vanishes in degrees greater than  $r$ . Formal functions shows that the completion of every stalk of  $R^i f_* \mathcal{F}$  is zero. Faithful flatness of completion then implies that the stalk itself is zero.  $\square$

**Remark 7.15.** Properness is essential. For  $j: \mathbb{A}_k^n \setminus \{0\} \hookrightarrow \mathbb{A}_k^n$ , one has

$$R^{n-1} j_* \mathcal{O}_{\mathbb{A}_k^n \setminus \{0\}} \neq 0.$$

## 7.6 Connected fibers and Stein factorization

**Definition 7.16.** A proper morphism  $f: X \rightarrow Y$  is  $\mathcal{O}$ -connected if

$$f_* \mathcal{O}_X = \mathcal{O}_Y.$$

**Theorem 7.17** (Zariski connectedness theorem). *Let  $f: X \rightarrow Y$  be a proper morphism of locally Noetherian schemes. If  $f_* \mathcal{O}_X = \mathcal{O}_Y$ , then every fiber of  $f$  is connected.*

*Proof.* If  $X_y = U \sqcup V$  were disconnected, its two components would define a nontrivial idempotent in  $H^0(X_y, \mathcal{O}_{X_y})$ . Idempotents lift uniquely through nilpotent thickenings, so this gives a nontrivial idempotent in

$$\varprojlim_n H^0(X_n, \mathcal{O}_{X_n}).$$

By formal functions and the hypothesis, this inverse limit is  $\widehat{\mathcal{O}}_{Y,y}$ , which is local and hence has no nontrivial idempotents. This is a contradiction.  $\square$

**Theorem 7.18** (Stein factorization). *Let  $f: X \rightarrow Y$  be proper, where  $Y$  is locally Noetherian. Then  $f$  factors canonically as*

$$X \xrightarrow{g} Y' \xrightarrow{\pi} Y, \quad Y' = \underline{\mathrm{Spec}}_Y(f_* \mathcal{O}_X),$$

where  $g$  is proper and  $\mathcal{O}$ -connected, and  $\pi$  is finite. In particular,  $g$  has connected fibers.

**Corollary 7.19** (Zariski's connectedness theorem for birational maps). *Let  $f: X \rightarrow Y$  be a proper birational morphism of integral varieties. If  $Y$  is normal, then*

$$f_* \mathcal{O}_X = \mathcal{O}_Y,$$

and all fibers of  $f$  are connected.

*Proof.* Apply Stein factorization. The finite map  $\pi: Y' \rightarrow Y$  is birational. On an affine open  $\mathrm{Spec} A \subseteq Y$ , it corresponds to a finite intermediate algebra

$$A \subseteq B \subseteq K(Y).$$

Since  $A$  is integrally closed,  $B = A$ . Thus  $\pi$  is an isomorphism.  $\square$

### 7.6.1 An application to a blow-up

Let  $X$  be smooth of dimension  $d$ , let  $x \in X$ , and let  $\beta: \tilde{X} = \text{Bl}_x(X) \rightarrow X$ . Then

$$\beta_* \mathcal{O}_{\tilde{X}} = \mathcal{O}_X, \quad R^i \beta_* \mathcal{O}_{\tilde{X}} = 0 \quad (i > 0).$$

The first assertion follows from normality and the preceding corollary. For the second, formal functions reduces the calculation at  $x$  to the infinitesimal neighborhoods of

$$E \simeq \mathbb{P}_k^{d-1}.$$

Their successive ideal quotients are nonnegative twists on  $E$ , and

$$H^i(\mathbb{P}_k^{d-1}, \mathcal{O}(n)) = 0 \quad (i > 0, n \geq 0).$$

It follows inductively that all completed stalks of the higher direct images vanish.

The Leray spectral sequence therefore gives

$$H^i(\tilde{X}, \mathcal{O}_{\tilde{X}}) \simeq H^i(X, \mathcal{O}_X) \quad (i \geq 0).$$

This does not mean that all cohomology is unchanged. For example, if  $\tilde{X} = \text{Bl}_p(\mathbb{P}_{\mathbb{C}}^2)$ , then

$$H^2(\mathbb{P}_{\mathbb{C}}^2, \mathbb{Z}) \simeq \mathbb{Z}, \quad H^2(\tilde{X}, \mathbb{Z}) \simeq \mathbb{Z}^2.$$

The additional class is represented by the exceptional curve  $E$ ; indeed

$$\text{Pic}(\tilde{X}) \simeq \mathbb{Z}[H] \oplus \mathbb{Z}[E], \quad \mathcal{O}_E(E) \simeq \mathcal{O}_E(-1).$$

## 7.7 Isolated points and Zariski's Main Theorem

**Definition 7.20.** Let  $f: X \rightarrow Y$  be proper and let  $x \in X$ , with  $y = f(x)$ . The point  $x$  is *isolated in its fiber* if  $\{x\}$  is a connected component of  $X_y$ .

**Proposition 7.21.** Let  $f: X \rightarrow Y$  be proper and  $\mathcal{O}$ -connected.

1. The morphism  $f$  is surjective.
2. A point  $x \in X$  is isolated in its fiber if and only if  $f$  is unramified at  $x$ . On this locus,  $f$  is locally an isomorphism.

*Proof.* The image of  $f$  is closed, while the equality  $f_* \mathcal{O}_X = \mathcal{O}_Y$  forces it to be schematically dense; hence it is all of  $Y$ . If  $f$  is unramified at  $x$ , then it is locally quasi-finite, so  $x$  is isolated in the fiber. Conversely, connectedness of the fibers shows that an isolated point is the entire corresponding fiber. Properness allows one to shrink the base so that its inverse image lies in any prescribed neighborhood of  $x$ . The equality of structure sheaves then makes the induced map on local rings an isomorphism.  $\square$

**Theorem 7.22** (Isolated-point form of Zariski's Main Theorem). Let  $f: X \rightarrow Y$  be proper and let

$$X \xrightarrow{g} Y' \xrightarrow{\pi} Y$$

be its Stein factorization. The set  $X_0 \subseteq X$  of points isolated in their fibers under  $g$  is open, and  $g$  restricts to an open immersion

$$X_0 \hookrightarrow Y'.$$

Thus

$$f|_{X_0}: X_0 \hookrightarrow Y' \xrightarrow{\pi} Y$$

is the composite of an open immersion and a finite morphism.

**Corollary 7.23.** *For a morphism  $f: X \rightarrow Y$  of locally Noetherian schemes, the following are equivalent:*

1.  $f$  is finite;
2.  $f$  is affine and proper;
3.  $f$  is proper and quasi-finite.

*Proof.* A finite morphism is affine and proper. If  $f$  is affine and proper, the coherent algebra  $f_*\mathcal{O}_X$  is finite over  $\mathcal{O}_Y$ , so  $f$  is finite. Finally, if  $f$  is proper and quasi-finite, every point is isolated in its fiber. The preceding factorization makes  $f$  an open immersion followed by a finite map; properness makes that open immersion closed, hence finite.  $\square$

## 7.8 Affine morphisms

**Theorem 7.24.** *For a morphism  $f: X \rightarrow S$ , the following are equivalent:*

1.  $f$  is affine;
2. there is a quasi-coherent  $\mathcal{O}_S$ -algebra  $\mathcal{A}$  such that

$$X \simeq \underline{\mathrm{Spec}}_S(\mathcal{A}).$$

In this case  $\mathcal{A} = f_*\mathcal{O}_X$ .

**Theorem 7.25** (Serre's criterion for affine morphisms). *Let  $f: X \rightarrow S$  be quasi-compact and separated. Then  $f$  is affine if and only if*

$$f_*: \mathrm{QCoh}(X) \longrightarrow \mathrm{QCoh}(S)$$

is exact. For an affine morphism,  $f_*$  preserves quasi-coherent sheaves.

## Chapter 8

# Cohomology and Base Change

### 8.1 The base-change problem

Let  $f: X \rightarrow Y$  be a proper morphism of Noetherian schemes and let  $\mathcal{F}$  be a coherent sheaf on  $X$ . The higher direct images  $R^i f_* \mathcal{F}$  are coherent. For a point  $y \in Y$ , there is a natural comparison map

$$\varphi_i(y): (R^i f_* \mathcal{F}) \otimes k(y) \longrightarrow H^i(X_y, \mathcal{F}_y),$$

where

$$X_y = X \times_Y \operatorname{Spec} k(y), \quad \mathcal{F}_y = \mathcal{F}|_{X_y}.$$

Unlike the theorem on formal functions, this map need not be an isomorphism.

More generally, consider a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{g} & Y. \end{array}$$

There is a canonical base-change morphism

$$g^* R^i f_* \mathcal{F} \longrightarrow R^i f'_* h^* \mathcal{F}.$$

The central question is to determine conditions under which this morphism is an isomorphism.

### 8.2 A finite complex computing fiber cohomology

**Theorem 8.1** (The base-change complex). *Let  $A$  be Noetherian, let  $f: X \rightarrow \operatorname{Spec} A$  be projective, and let  $\mathcal{F}$  be a coherent sheaf that is flat over  $A$ . There is a bounded complex*

$$K^\bullet: \quad 0 \longrightarrow K^0 \longrightarrow K^1 \longrightarrow \cdots \longrightarrow K^n \longrightarrow 0$$

*of finite projective  $A$ -modules such that, for every  $A$ -algebra  $B$ ,*

$$H^i(X_B, \mathcal{F}_B) \simeq H^i(K^\bullet \otimes_A B).$$

Here  $X_B = X \times_A \operatorname{Spec} B$  and  $\mathcal{F}_B = \mathcal{F} \otimes_A B$ .

This theorem packages all fibers into one finite algebraic object. After restricting to an affine open of a general base  $Y$ , the same construction applies locally on  $Y$ .

*Idea of proof.* Choose a finite affine open cover of  $X$ . Since  $X$  is separated, all finite intersections are affine, and the corresponding Čech complex  $C^\bullet$  computes the cohomology of  $\mathcal{F}$ . The terms of  $C^\bullet$  are generally not finite over  $A$ , but they are flat because  $\mathcal{F}$  is  $A$ -flat. Properness implies that the cohomology modules  $H^i(C^\bullet)$  are finite.

One constructs a bounded complex  $K^\bullet$  of finite free modules, except initially possibly in degree zero, together with a quasi-isomorphism

$$K^\bullet \longrightarrow C^\bullet.$$

The construction proceeds downward: finite generators for the cohomology and boundary modules are lifted degree by degree. The mapping cone is an acyclic bounded complex of flat modules. Its first kernel is flat, which proves that the remaining degree-zero term of  $K^\bullet$  is finite flat, hence projective.

An acyclic bounded complex of flat modules remains acyclic after tensoring with any  $A$ -module. Therefore

$$K^\bullet \otimes_A B \longrightarrow C^\bullet \otimes_A B$$

is still a quasi-isomorphism. The latter is the Čech complex for  $(X_B, \mathcal{F}_B)$ , proving the claim.  $\square$

*Remark 8.2.* If  $A \rightarrow B$  is flat, tensoring preserves quasi-isomorphisms without requiring the terms of the original complexes to be flat. The projectivity and flatness assumptions above are needed for arbitrary base change and for the finite projective model.

### 8.3 Semicontinuity and the Euler characteristic

**Theorem 8.3** (Upper semicontinuity). *Let  $f: X \rightarrow Y$  be projective, with  $Y$  Noetherian, and let  $\mathcal{F}$  be coherent and flat over  $Y$ . For every  $i$ , the function*

$$h_{\mathcal{F}}^i: Y \longrightarrow \mathbb{Z}_{\geq 0}, \quad y \longmapsto \dim_{k(y)} H^i(X_y, \mathcal{F}_y)$$

*is upper semicontinuous. Equivalently, for every integer  $c$ , the set*

$$\{y \in Y \mid h_{\mathcal{F}}^i(y) \geq c\}$$

*is closed.*

*Proof.* Work over  $Y = \operatorname{Spec} A$  and use the finite projective complex  $K^\bullet$ . Locally its terms are free. At  $y$ ,

$$h_{\mathcal{F}}^i(y) = \operatorname{rank} K^i - \operatorname{rank}(d^{i-1} \otimes k(y)) - \operatorname{rank}(d^i \otimes k(y)).$$

The rank of a matrix over a residue field is lower semicontinuous: the locus where it is less than  $c$  is cut out by its  $c \times c$  minors. The displayed expression is therefore upper semicontinuous.  $\square$

**Corollary 8.4.** *Under the same hypotheses, the Euler characteristic*

$$\chi(X_y, \mathcal{F}_y) = \sum_i (-1)^i h^i(X_y, \mathcal{F}_y)$$

*is locally constant on  $Y$ .*

*Proof.* The differential ranks cancel in the alternating sum, giving

$$\chi(X_y, \mathcal{F}_y) = \sum_i (-1)^i \operatorname{rank} K^i.$$

The ranks of finite projective modules are locally constant.  $\square$

**Example 8.5** (Why flatness matters). Let  $\beta: \operatorname{Bl}_0(\mathbb{A}^2) \rightarrow \mathbb{A}^2$  be the blow-up at the origin, with exceptional divisor  $E \simeq \mathbb{P}^1$ . Although  $\mathcal{O}(E)$  is invertible on the source, it is not flat over  $\mathbb{A}^2$ , because  $\beta$  is not flat. Its restriction to a nonexceptional fiber is a one-dimensional vector space, whereas

$$\mathcal{O}(E)|_E \simeq \mathcal{O}_{\mathbb{P}^1}(-1).$$

Thus

$$h^0(\mathcal{O}(E)|_{\beta^{-1}(y)}) = \begin{cases} 1, & y \neq 0, \\ 0, & y = 0, \end{cases}$$

which is not upper semicontinuous.

Likewise, for  $f = \operatorname{id}_Y$  and  $\mathcal{F} = k(y_0)$ , the Euler characteristic of the fibers is zero away from  $y_0$  and one at  $y_0$ . This sheaf is not flat over  $Y$ .

## 8.4 The cohomology-and-base-change criterion

**Theorem 8.6** (Cohomology and base change). *Let  $f: X \rightarrow Y$  be projective, let  $Y$  be Noetherian, and let  $\mathcal{F}$  be coherent and flat over  $Y$ . Fix  $y \in Y$ .*

1. *If*

$$\varphi_i(y): (R^i f_* \mathcal{F}) \otimes k(y) \longrightarrow H^i(X_y, \mathcal{F}_y)$$

*is surjective, then it is an isomorphism. Moreover,  $\varphi_i$  remains an isomorphism in a neighborhood of  $y$ .*

2. *If  $\varphi_i(y)$  is surjective, then the following are equivalent:*

- (a)  $\varphi_{i-1}(y)$  is surjective;
- (b)  $R^i f_* \mathcal{F}$  is locally free in a neighborhood of  $y$ .

*Linear-algebra description.* Locally, replace  $Rf_* \mathcal{F}$  by  $K^\bullet$ . After choosing splittings around  $y$ , the condition that  $\varphi_i(y)$  be surjective says that the relevant boundary submodule has the expected rank after reduction modulo the maximal ideal. Nakayama's lemma then splits the corresponding differential locally. Cohomology therefore commutes with tensor product in degree  $i$ , proving the first assertion. The preceding differential splits as well precisely when  $H^i(K^\bullet) = R^i f_* \mathcal{F}$  is locally free, which gives the equivalence in the second assertion.  $\square$



**Corollary 8.7.** *If  $H^i(X_y, \mathcal{F}_y) = 0$ , then the natural map*

$$(R^{i-1}f_*\mathcal{F}) \otimes k(y) \xrightarrow{\sim} H^{i-1}(X_y, \mathcal{F}_y)$$

*is an isomorphism.*

**Corollary 8.8.** *If*

$$R^i f_* \mathcal{F} = 0 \quad (i \geq i_0),$$

*then*

$$H^i(X_y, \mathcal{F}_y) = 0 \quad (i \geq i_0, y \in Y).$$

*Proof.* Fiber cohomology vanishes in sufficiently high degrees. Descend on  $i$ . If the fiber cohomology vanishes in degree  $i + 1$ , the preceding corollary identifies degree  $i$  fiber cohomology with  $(R^i f_* \mathcal{F}) \otimes k(y)$ , which is zero.  $\square$

**Theorem 8.9** (Grauert-type criterion). *Let  $f: X \rightarrow Y$  be projective, let  $Y$  be reduced and Noetherian, and let  $\mathcal{F}$  be coherent and flat over  $Y$ . For a fixed  $i$ , the following are equivalent:*

1. *the function  $y \mapsto h^i(X_y, \mathcal{F}_y)$  is locally constant;*
2.  *$R^i f_* \mathcal{F}$  is locally free and every base-change map*

$$(R^i f_* \mathcal{F}) \otimes k(y) \xrightarrow{\sim} H^i(X_y, \mathcal{F}_y)$$

*is an isomorphism.*

*Sketch.* The reverse implication is immediate. For the forward implication, use the finite projective complex. Constancy of the cohomology dimension and semicontinuity of the ranks of the adjacent differentials force those ranks to be locally constant. A map of finite locally free modules having constant rank over a reduced base is locally equivalent to a block matrix with an identity block and a zero block. Thus the relevant differentials split locally. Consequently  $H^i(K^\bullet)$  is locally free and commutes with tensoring by residue fields.  $\square$

## 8.5 Flat base change

**Theorem 8.10** (Flat base change). *In a Cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{g} & Y, \end{array}$$

*assume that  $f$  is proper between Noetherian schemes and that  $g$  is flat. Then, for every coherent sheaf  $\mathcal{F}$ ,*

$$g^* R^i f_* \mathcal{F} \xrightarrow{\sim} R^i f'_* h^* \mathcal{F}.$$

*Proof.* The statement is local on  $Y$  and  $Y'$ . Write  $Y = \operatorname{Spec} A$  and  $Y' = \operatorname{Spec} B$ . Since  $B$  is flat over  $A$ , cohomology commutes with tensoring by  $B$ :

$$H^i(X, \mathcal{F}) \otimes_A B \xrightarrow{\sim} H^i(X_B, \mathcal{F}_B).$$

Sheafifying this isomorphism gives the result.  $\square$

*Remark 8.11.* The direction of the natural base-change morphism can be remembered from the adjunction map  $\mathcal{F} \rightarrow h_* h^* \mathcal{F}$ . Applying  $f_*$ , using  $f \circ h = g \circ f'$ , and then adjunction for  $g^* \dashv g_*$  produces

$$g^* f_* \mathcal{F} \longrightarrow f'_* h^* \mathcal{F},$$

and similarly on higher direct images.

## 8.6 Triviality loci and the see-saw principle

**Proposition 8.12** (Closedness of the triviality locus). *Let  $f: X \rightarrow T$  be flat and projective with geometrically integral fibers, and let  $\mathcal{L}$  be a line bundle on  $X$ . Then*

$$Z(\mathcal{L}) = \{t \in T \mid \mathcal{L}_t \simeq \mathcal{O}_{X_t}\}$$

*is closed in  $T$ .*

*Proof.* On a proper integral scheme over a field, a line bundle  $L$  is trivial if and only if both  $L$  and  $L^\vee$  have nonzero global sections. Indeed, the product of two such sections is a nonzero global function, hence a unit, so both sections are nowhere vanishing. Therefore

$$Z(\mathcal{L}) = \{t \mid h^0(X_t, \mathcal{L}_t) \geq 1\} \cap \{t \mid h^0(X_t, \mathcal{L}_t^\vee) \geq 1\},$$

which is closed by upper semicontinuity.  $\square$

**Theorem 8.13** (Descent of a fiberwise trivial line bundle). *Let  $f: X \rightarrow Y$  be flat, projective, and surjective, with geometrically integral fibers, and assume that  $Y$  is reduced and Noetherian. If a line bundle  $\mathcal{L}$  on  $X$  satisfies*

$$\mathcal{L}_y \simeq \mathcal{O}_{X_y} \quad \text{for every } y \in Y,$$

*then there is a line bundle  $M$  on  $Y$  such that*

$$\mathcal{L} \simeq f^* M.$$

*Proof.* Every fiber has  $h^0(X_y, \mathcal{L}_y) = 1$ . The Grauert-type criterion shows that

$$M = f_* \mathcal{L}$$

is a line bundle and that its formation commutes with passage to every fiber. The adjunction morphism

$$f^* M \longrightarrow \mathcal{L}$$

is an isomorphism on every fiber, hence an isomorphism.  $\square$

**Corollary 8.14** (See-saw principle). *Let  $Z$  be a geometrically integral projective  $k$ -scheme, let  $Y$  be reduced and Noetherian, and let  $\mathcal{L}_1, \mathcal{L}_2 \in \text{Pic}(Z \times Y)$ . Suppose*

$$\mathcal{L}_1|_{Z \times \{y\}} \simeq \mathcal{L}_2|_{Z \times \{y\}} \quad \text{for every } y \in Y.$$

Then there is  $M \in \text{Pic}(Y)$  such that

$$\mathcal{L}_1 \simeq \mathcal{L}_2 \otimes \text{pr}_Y^* M.$$

If, in addition, the two line bundles have isomorphic restrictions to  $\{z\} \times Y$  for some  $z \in Z(k)$ , then  $M \simeq \mathcal{O}_Y$  and  $\mathcal{L}_1 \simeq \mathcal{L}_2$ .

## 8.7 Openness of ampleness

**Theorem 8.15.** *Let  $f: X \rightarrow Y$  be projective, with  $Y$  Noetherian, and let  $\mathcal{L}$  be a line bundle on  $X$ . The fiberwise ampleness locus*

$$A_{\mathcal{L}} = \{y \in Y \mid \mathcal{L}_y \text{ is ample on } X_y\}$$

*is open in  $Y$ .*

*Idea.* If  $\mathcal{L}_y$  is ample, a sufficiently high power has enough sections to define a closed immersion of  $X_y$  into projective space, with vanishing higher cohomology for the relevant twists. Upper semicontinuity and cohomology and base change extend these properties to a neighborhood of  $y$ . The resulting relative sections define a closed immersion near the fiber, proving relative, and hence fiberwise, ampleness on that neighborhood.  $\square$

## 8.8 A failure of base change

Let  $E$  be an elliptic curve over  $k$ , put  $X = E \times E$ , and let  $f = \text{pr}_1$ . Denote by  $\Delta$  the diagonal and by  $E \times \{0\}$  the horizontal divisor. Consider

$$\mathcal{L} = \mathcal{O}_X(\Delta - E \times \{0\}).$$

On the fiber  $\{y\} \times E$ ,

$$\mathcal{L}_y \simeq \mathcal{O}_E(y - 0).$$

For  $y \neq 0$ , this is a nontrivial degree-zero line bundle, so

$$H^0(E, \mathcal{L}_y) = 0.$$

At  $y = 0$ , however,  $\mathcal{L}_0 \simeq \mathcal{O}_E$ , and hence

$$H^0(E, \mathcal{L}_0) \simeq k.$$

The sheaf  $f_*\mathcal{L}$  vanishes: it is torsion-free on the smooth curve  $E$ , while its generic rank is zero. Consequently

$$(f_*\mathcal{L}) \otimes k(0) = 0 \quad \text{but} \quad H^0(E, \mathcal{L}_0) = k.$$

Thus the base-change map in degree zero is not an isomorphism. This example also illustrates upper semicontinuity: the dimension of global sections jumps upward at the origin.

## Chapter 9

# Abelian Varieties

### 9.1 Definitions and translations

**Definition 9.1.** An *abelian variety* over a field  $k$  is a proper, smooth, geometrically connected group scheme of finite type over  $k$ .

Thus an abelian variety  $A$  is equipped with morphisms

$$\mu: A \times A \longrightarrow A, \quad \iota: A \longrightarrow A, \quad e: \operatorname{Spec} k \longrightarrow A$$

satisfying the group axioms. The multiplication is associative:

$$\begin{array}{ccc} A \times A \times A & \xrightarrow{\mu \times \operatorname{id}} & A \times A \\ \operatorname{id} \times \mu \downarrow & & \downarrow \mu \\ A \times A & \xrightarrow{\mu} & A. \end{array}$$

The morphism  $e$  is a two-sided identity, and

$$\mu \circ (\operatorname{id}, \iota) \circ \Delta = \mu \circ (\iota, \operatorname{id}) \circ \Delta = e.$$

The group schemes  $\mathbb{G}_a$ ,  $\mathbb{G}_m$ , and  $\operatorname{GL}_n$  are not abelian varieties because they are not proper.

*Remark 9.2.* Some texts begin with a proper group variety and deduce smoothness. This is automatic over a perfect field. Including smoothness and geometric connectedness in the definition gives the standard field-independent formulation.

For  $x \in A(k)$ , translation by  $x$  is

$$t_x: A \longrightarrow A, \quad y \longmapsto y + x.$$

Scheme-theoretically, it is the composite

$$A \xrightarrow{(\operatorname{id}, x)} A \times A \xrightarrow{\mu} A.$$

Translations satisfy

$$t_y \circ t_x = t_{x+y}, \quad t_x^{-1} = t_{-x}.$$

More generally, if  $S$  is a  $k$ -scheme and  $\tau \in A(S)$ , then translation by  $\tau$  is an  $S$ -automorphism of  $A_S = A \times_k S$ .

## 9.2 Invariant differential forms

**Proposition 9.3.** *Let  $A$  be an abelian variety. Translation identifies the tangent bundle with the constant vector bundle determined by the tangent space at the identity:*

$$\mathcal{T}_{A/k} \simeq T_{A,e} \otimes_k \mathcal{O}_A.$$

Consequently,

$$\Omega_{A/k}^1 \simeq T_{A,e}^\vee \otimes_k \mathcal{O}_A, \quad \Omega_{A/k}^p \simeq \bigwedge^p T_{A,e}^\vee \otimes_k \mathcal{O}_A.$$

In particular,

$$\omega_A \simeq \mathcal{O}_A.$$

*Proof.* The differential of the group law along  $A \times \{e\}$  transports each tangent vector at  $e$  to a translation-invariant vector field on  $A$ . This gives a morphism of vector bundles

$$T_{A,e} \otimes_k \mathcal{O}_A \longrightarrow \mathcal{T}_{A/k}.$$

Its fiber at  $x \in A$  is the differential of  $t_x$  at  $e$ , and is therefore an isomorphism. Hence the morphism is an isomorphism everywhere. Dualizing and taking exterior powers gives the remaining statements.  $\square$

Thus every global differential form obtained from a cotangent vector at  $e$  is invariant under all translations.

## 9.3 Rigidity and morphisms of abelian varieties

**Theorem 9.4** (Rigidity lemma). *Let  $X$  be a proper geometrically connected variety, let  $Y$  be a connected variety, and let  $Z$  be a separated variety. If a morphism*

$$F: X \times Y \longrightarrow Z$$

*maps  $X \times \{y_0\}$  to a point for some  $y_0 \in Y$ , then  $F$  factors through the second projection:*

$$F = G \circ \text{pr}_2$$

*for a unique morphism  $G: Y \rightarrow Z$ .*

**Proposition 9.5.** *Let  $A$  and  $B$  be abelian varieties. Every morphism of varieties  $f: A \rightarrow B$  can be written uniquely as*

$$f = t_{f(e)} \circ h,$$

*where  $h: A \rightarrow B$  is a homomorphism of algebraic groups.*

*Proof.* Put  $h = t_{-f(e)} \circ f$ , so that  $h(e) = e$ , and consider

$$G: A \times A \longrightarrow B, \quad G(x, y) = h(x + y) - h(x) - h(y).$$

The restriction of  $G$  to  $A \times \{e\}$  is constant. Rigidity shows that  $G$  factors through the second projection. Since its restriction to  $\{e\} \times A$  is also constant,  $G$  is the constant map with value  $e$ . Thus

$$h(x + y) = h(x) + h(y),$$

so  $h$  is a group homomorphism. Its definition makes the factorization unique.  $\square$

**Corollary 9.6.**

1. On a fixed proper variety  $A$ , there is at most one abelian variety group law having a prescribed point  $e$  as identity.
2. Every abelian variety is commutative.

*Proof.* For the first assertion, apply the proposition to the identity map between the two proposed group structures. For the second, inversion  $\iota: A \rightarrow A$  fixes the identity, so the proposition makes it a homomorphism. On the other hand, the group axioms give

$$\iota(x + y) = \iota(y) + \iota(x).$$

Comparison with  $\iota(x + y) = \iota(x) + \iota(y)$  yields  $x + y = y + x$ .  $\square$

**Example 9.7** (Elliptic curves). Let  $E \subseteq \mathbb{P}_k^2$  be a smooth plane cubic with a chosen point  $e \in E(k)$ . It has genus one and trivial canonical bundle, and the choice of  $e$  determines an abelian variety structure. Geometrically, let a line meet  $E$  in  $P, Q, R$ , counted with multiplicity, and let the line through  $R$  and  $e$  meet  $E$  again at  $S$ . Then  $P + Q = S$ . If  $e$  is a flex, this reduces to the familiar rule that three collinear points satisfy

$$P + Q + R = e.$$

## 9.4 The see-saw theorem

**Theorem 9.8** (See-saw). Let  $X$  be a proper geometrically integral  $k$ -variety and let  $Y$  be a reduced  $k$ -variety. Suppose  $\mathcal{L}, \mathcal{M} \in \text{Pic}(X \times Y)$  satisfy

$$\mathcal{L}|_{X \times \{y\}} \simeq \mathcal{M}|_{X \times \{y\}} \quad \text{for every closed point } y \in Y.$$

Then there is a line bundle  $N \in \text{Pic}(Y)$  such that

$$\mathcal{L} \simeq \mathcal{M} \otimes \text{pr}_2^* N.$$

If their restrictions to  $\{x\} \times Y$  are also isomorphic for some  $x \in X(k)$ , then  $\mathcal{L} \simeq \mathcal{M}$ .

*Proof.* Apply the descent result from cohomology and base change to  $\mathcal{L} \otimes \mathcal{M}^\vee$ , which is trivial on every fiber of  $\text{pr}_2$ . It is therefore pulled back from a line bundle  $N$  on  $Y$ . Restriction to  $\{x\} \times Y$  identifies that pullback with  $N$ , proving the final assertion.  $\square$

**Proposition 9.9.** *Let  $X$  be proper and geometrically integral, let  $Y$  be a variety, and let  $\mathcal{L} \in \text{Pic}(X \times Y)$ . Then*

$$\{y \in Y \mid \mathcal{L}|_{X \times \{y\}} \simeq \mathcal{O}_{X_y}\}$$

*is closed.*

*Proof.* For a line bundle  $M$  on a proper integral variety,

$$M \simeq \mathcal{O} \iff H^0(M) \neq 0 \text{ and } H^0(M^\vee) \neq 0.$$

The result follows by upper semicontinuity applied to  $\mathcal{L}$  and  $\mathcal{L}^\vee$ .  $\square$

## 9.5 The theorem of the cube

**Theorem 9.10** (The theorem of the cube, geometric form). *Let  $X$  and  $Y$  be proper geometrically integral  $k$ -varieties, let  $Z$  be a connected locally Noetherian  $k$ -scheme, and choose  $x \in X(k)$ ,  $y \in Y(k)$ , and  $z \in Z(k)$ . Let  $\mathcal{L}$  be a line bundle on  $X \times Y \times Z$ . If*

$$\mathcal{L}|_{\{x\} \times Y \times Z} \simeq \mathcal{O},$$

$$\mathcal{L}|_{X \times \{y\} \times Z} \simeq \mathcal{O},$$

$$\mathcal{L}|_{X \times Y \times \{z\}} \simeq \mathcal{O},$$

*then  $\mathcal{L} \simeq \mathcal{O}_{X \times Y \times Z}$ .*

*Idea of proof.* The locus in  $Z$  on which the restriction of  $\mathcal{L}$  to  $X \times Y$  is trivial is closed and contains  $z$ . It is also open. To see this, consider a square-zero infinitesimal extension  $Z_1 \hookrightarrow Z_2$ . The obstruction to lifting a trivialization belongs to

$$H^1(X \times Y, \mathcal{O}_{X \times Y}) \simeq H^1(X, \mathcal{O}_X) \oplus H^1(Y, \mathcal{O}_Y).$$

The first two normalization conditions make both components vanish, so the trivialization lifts. Applying this successively to Artinian thickenings and using Krull intersection shows that the triviality locus is open. Since  $Z$  is connected, it is all of  $Z$ .  $\square$

*Remark 9.11.* In general,

$$\text{Pic}(X \times Y) \neq \text{Pic}(X) \oplus \text{Pic}(Y).$$

For an elliptic curve  $E$ , the divisor

$$\Delta - \{e\} \times E - E \times \{e\}$$

on  $E \times E$  restricts trivially to both coordinate axes but is not itself linearly equivalent to zero.

### 9.5.1 The cubical identity on an abelian variety

Let  $A$  be an abelian variety. For  $I \subseteq \{1, 2, 3\}$ , write

$$p_I: A^3 \longrightarrow A, \quad (x_1, x_2, x_3) \longmapsto \sum_{i \in I} x_i.$$

**Theorem 9.12** (The theorem of the cube). *For every  $\mathcal{L} \in \text{Pic}(A)$ , the line bundle*

$$\begin{aligned} \Theta(\mathcal{L}) = & p_{123}^* \mathcal{L} \otimes p_{12}^* \mathcal{L}^\vee \otimes p_{13}^* \mathcal{L}^\vee \otimes p_{23}^* \mathcal{L}^\vee \\ & \otimes p_1^* \mathcal{L} \otimes p_2^* \mathcal{L} \otimes p_3^* \mathcal{L} \end{aligned}$$

*is trivial.*

*Proof.* Restricting  $\Theta(\mathcal{L})$  to any coordinate plane on which one variable is  $e$  makes all factors cancel, up to a constant one-dimensional  $k$ -vector space. The geometric form of the theorem of the cube then gives the result.  $\square$

**Corollary 9.13.** *For a scheme  $S$  and morphisms  $f, g, h: S \rightarrow A$ , the line bundle*

$$\begin{aligned} & (f + g + h)^* \mathcal{L} \otimes (f + g)^* \mathcal{L}^\vee \otimes (f + h)^* \mathcal{L}^\vee \otimes (g + h)^* \mathcal{L}^\vee \\ & \otimes f^* \mathcal{L} \otimes g^* \mathcal{L} \otimes h^* \mathcal{L} \end{aligned}$$

*is trivial.*

## 9.6 The homomorphism associated with a line bundle

The theorem of the cube implies the theorem of the square:

$$t_{x+y}^* \mathcal{L} \otimes \mathcal{L}^\vee \simeq (t_x^* \mathcal{L} \otimes \mathcal{L}^\vee) \otimes (t_y^* \mathcal{L} \otimes \mathcal{L}^\vee).$$

Consequently,

$$\varphi_{\mathcal{L}}: A(k) \longrightarrow \text{Pic}(A), \quad x \longmapsto t_x^* \mathcal{L} \otimes \mathcal{L}^\vee$$

is a group homomorphism. In fact its image lies in  $\text{Pic}^0(A)$ , and the construction is induced by a morphism of abelian varieties

$$\varphi_{\mathcal{L}}: A \longrightarrow A^\vee,$$

where  $A^\vee = \text{Pic}_{A/k}^0$  is the dual abelian variety.

**Example 9.14.** Let  $E$  be an elliptic curve with identity  $e$ , and take  $\mathcal{L} = \mathcal{O}_E(e)$ . With the convention  $t_x(y) = y + x$ ,

$$t_x^* \mathcal{O}_E(e) \otimes \mathcal{O}_E(-e) \simeq \mathcal{O}_E((-x) - (e)).$$

Thus  $\varphi_{\mathcal{L}}$  is the standard identification

$$E \xrightarrow{\sim} \text{Pic}^0(E)$$

composed with inversion. In particular, it is an isomorphism.



## Chapter 10

# Serre Duality and Derived Methods

### 10.1 A collection of cohomological formulas

Before proving Serre duality, we record several formulas that place it in context.

#### 10.1.1 Riemann–Roch and Riemann–Hurwitz

Let  $C$  be a smooth projective curve of genus  $g$ . For a line bundle  $L$ ,

$$\chi(C, L) = \deg L + 1 - g.$$

More generally, for a vector bundle  $E$ ,

$$\chi(C, E) = \deg E + \operatorname{rank}(E)(1 - g).$$

For a smooth projective variety  $X$ , Hirzebruch–Riemann–Roch takes the form

$$\chi(X, E) = \int_X \operatorname{ch}(E) \operatorname{td}(T_X).$$

The relative form is Grothendieck–Riemann–Roch. If  $f: X \rightarrow Y$  is a proper morphism between smooth varieties, then

$$f_*(\operatorname{ch}(E) \operatorname{td}(T_X)) = \operatorname{ch}(Rf_*E) \operatorname{td}(T_Y).$$

If  $f: C_1 \rightarrow C_2$  is a finite separable morphism of smooth projective curves, then

$$2g(C_1) - 2 = \deg(f)(2g(C_2) - 2) + \deg R_f,$$

where  $R_f$  is the ramification, or different, divisor.

### 10.1.2 Projection and Künneth formulas

For a proper morphism  $f: X \rightarrow Y$ , a complex  $\mathcal{F}^\bullet$  on  $X$ , and a perfect complex  $\mathcal{G}^\bullet$  on  $Y$ , the derived projection formula is

$$Rf_*(\mathcal{F}^\bullet \otimes^L Lf^*\mathcal{G}^\bullet) \simeq Rf_*\mathcal{F}^\bullet \otimes^L \mathcal{G}^\bullet.$$

For a locally free sheaf  $\mathcal{G}$ , this gives

$$R^i f_*(\mathcal{F} \otimes f^*\mathcal{G}) \simeq R^i f_*\mathcal{F} \otimes \mathcal{G}.$$

If  $X_1, X_2$  are proper over  $k$ , then for coherent sheaves  $\mathcal{F}_i$  there is a Künneth isomorphism

$$H^n(X_1 \times X_2, \mathcal{F}_1 \boxtimes \mathcal{F}_2) \simeq \bigoplus_{p+q=n} H^p(X_1, \mathcal{F}_1) \otimes_k H^q(X_2, \mathcal{F}_2).$$

### 10.1.3 Hodge numbers and vanishing

Let  $X$  be smooth projective of dimension  $n$ , and put

$$h^{p,q}(X) = \dim_k H^q(X, \Omega_X^p).$$

Serre duality and the perfect wedge pairing

$$\Omega_X^p \otimes \Omega_X^{n-p} \longrightarrow \omega_X$$

give

$$h^{p,q}(X) = h^{n-p, n-q}(X).$$

Over a field of characteristic zero, Hodge theory also gives

$$h^{p,q}(X) = h^{q,p}(X).$$

This second symmetry can fail in positive characteristic.

The Kodaira–Akizuki–Nakano vanishing theorem states that, in characteristic zero, for an ample line bundle  $L$ ,

$$H^q(X, \Omega_X^p \otimes L) = 0 \quad \text{when } p + q > n.$$

In positive characteristic such a statement is false without additional hypotheses.

### 10.1.4 Fixed-point formulas

Let  $u: X \rightarrow X$  be an automorphism of a smooth projective variety, and suppose its fixed points are isolated and nondegenerate. The coherent Lefschetz formula is

$$\sum_{x \in \text{Fix}(u)} \frac{1}{\det(1 - du_x)} = \sum_i (-1)^i \text{tr}(u^* | H^i(X, \mathcal{O}_X)).$$

For  $X/\mathbb{F}_q$ , the  $\ell$ -adic trace formula gives

$$\#X(\mathbb{F}_q) = \sum_i (-1)^i \operatorname{tr} \left( \operatorname{Frob}_q^* \mid H_{\text{ét}}^i(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell) \right).$$

## 10.2 Classical Serre duality

Let  $X$  be a smooth projective variety of dimension  $n$  over  $k$ , and let

$$\omega_X = \bigwedge^n \Omega_{X/k}^1.$$

**Theorem 10.1** (Serre duality). *For every coherent sheaf  $\mathcal{F}$  on  $X$ , there are natural isomorphisms*

$$\operatorname{Ext}_X^i(\mathcal{F}, \omega_X) \simeq H^{n-i}(X, \mathcal{F})^\vee.$$

More generally, for coherent sheaves  $\mathcal{F}, \mathcal{G}$ ,

$$\operatorname{Ext}_X^i(\mathcal{F}, \mathcal{G} \otimes \omega_X) \simeq \operatorname{Ext}_X^{n-i}(\mathcal{G}, \mathcal{F})^\vee.$$

If  $\mathcal{F}$  is locally free, the first formula becomes

$$H^i(X, \mathcal{F}^\vee \otimes \omega_X) \simeq H^{n-i}(X, \mathcal{F})^\vee.$$

The duality is induced by the nondegenerate pairing

$$\begin{aligned} H^i(X, \mathcal{F}^\vee \otimes \omega_X) \times H^{n-i}(X, \mathcal{F}) &\longrightarrow H^n(X, \omega_X) \\ &\xrightarrow{\operatorname{tr}_X} k. \end{aligned}$$

## 10.3 Dualizing sheaves

**Definition 10.2.** Let  $X$  be a projective scheme of pure dimension  $n$ . A *dualizing sheaf* is a coherent sheaf  $\omega_X$ , together with a trace map

$$\operatorname{tr}_X: H^n(X, \omega_X) \longrightarrow k,$$

such that, for every coherent sheaf  $\mathcal{F}$ , the pairing

$$\operatorname{Hom}_X(\mathcal{F}, \omega_X) \times H^n(X, \mathcal{F}) \longrightarrow H^n(X, \omega_X) \xrightarrow{\operatorname{tr}_X} k$$

is perfect.

Equivalently,

$$\operatorname{Hom}_X(\mathcal{F}, \omega_X) \simeq H^n(X, \mathcal{F})^\vee$$

naturally in  $\mathcal{F}$ . Such a sheaf is unique up to unique isomorphism compatible with the trace.

### 10.3.1 Construction by an embedding

Choose a closed immersion

$$i: X \hookrightarrow \mathbb{P}_k^N,$$

and put  $c = N - n$ . Write  $\omega_{\mathbb{P}^N} = \mathcal{O}_{\mathbb{P}^N}(-N - 1)$ .

**Proposition 10.3.** *The sheaf*

$$\omega_X = \mathcal{E}xt_{\mathbb{P}^N}^c(\mathcal{O}_X, \omega_{\mathbb{P}^N})$$

*is a dualizing sheaf on  $X$ .*

*Outline.* One first proves

$$\mathcal{E}xt_{\mathbb{P}^N}^j(\mathcal{O}_X, \omega_{\mathbb{P}^N}) = 0 \quad (j < c).$$

After a sufficiently positive twist, the local-to-global Ext spectral sequence degenerates by Serre vanishing. Duality on projective space then identifies the relevant global Ext groups with the duals of top-degree cohomology on  $X$ .

The change-of-rings spectral sequence consequently gives

$$\mathrm{Hom}_X(\mathcal{F}, \omega_X) \simeq \mathrm{Ext}_{\mathbb{P}^N}^c(\mathcal{F}, \omega_{\mathbb{P}^N}) \simeq H^n(X, \mathcal{F})^\vee.$$

Taking  $\mathcal{F} = \omega_X$ , the identity endomorphism determines the trace map. □

*Remark 10.4.* For a general projective scheme, the full duality theory is naturally expressed using a dualizing complex. If  $X$  is Cohen–Macaulay, the dualizing complex is a shift of the dualizing sheaf, and classical Serre duality has the same form as above. The dualizing sheaf need not be invertible unless  $X$  is Gorenstein.

### 10.3.2 The smooth case and adjunction

**Proposition 10.5.** *If  $X$  is smooth and projective, then the dualizing sheaf constructed above is canonically isomorphic to*

$$\det \Omega_{X/k}^1.$$

*Proof.* The immersion  $X \hookrightarrow \mathbb{P}^N$  is regular. Locally its ideal is generated by a regular sequence of length  $c$ , so the Koszul complex is a locally free resolution of  $\mathcal{O}_X$ . Applying  $\mathcal{H}om_{\mathbb{P}^N}(-, \omega_{\mathbb{P}^N})$  gives

$$\mathcal{E}xt_{\mathbb{P}^N}^c(\mathcal{O}_X, \omega_{\mathbb{P}^N}) \simeq \omega_{\mathbb{P}^N}|_X \otimes \det N_{X/\mathbb{P}^N}.$$

Taking determinants in the conormal sequence

$$0 \longrightarrow N_{X/\mathbb{P}^N}^\vee \longrightarrow \Omega_{\mathbb{P}^N}^1|_X \longrightarrow \Omega_X^1 \longrightarrow 0$$

identifies the right-hand side with  $\det \Omega_X^1$ . □

## 10.4 Proof strategy for Serre duality

### 10.4.1 Projective space

On  $\mathbb{P}^n$ ,

$$\omega_{\mathbb{P}^n} = \mathcal{O}_{\mathbb{P}^n}(-n-1), \quad H^n(\mathbb{P}^n, \omega_{\mathbb{P}^n}) \simeq k.$$

For every integer  $a$ , multiplication gives a perfect pairing

$$H^0(\mathbb{P}^n, \mathcal{O}(-n-1-a)) \times H^n(\mathbb{P}^n, \mathcal{O}(a)) \longrightarrow k.$$

A presentation of a coherent sheaf by sums of line bundles extends this result to

$$\mathrm{Hom}(\mathcal{F}, \omega_{\mathbb{P}^n}) \simeq H^n(\mathbb{P}^n, \mathcal{F})^\vee.$$

The two families

$$\{\mathrm{Ext}^i(-, \omega_{\mathbb{P}^n})\} \quad \text{and} \quad \{H^{n-i}(\mathbb{P}^n, -)^\vee\}$$

are universal cohomological  $\delta$ -functors. Resolving by sufficiently negative line bundles and using the cohomology of  $\mathcal{O}(a)$  proves that they agree in every degree.

### 10.4.2 Passage to a smooth projective variety

For a smooth projective  $X$ , comparison maps

$$\theta_{\mathcal{F}}^i: \mathrm{Ext}_X^i(\mathcal{F}, \omega_X) \longrightarrow H^{n-i}(X, \mathcal{F})^\vee$$

are obtained from the degree-zero duality by the universal property of  $\delta$ -functors. Smoothness implies that  $X$  is Cohen–Macaulay. Equivalently, locally over  $\mathbb{P}^N$ ,  $\mathcal{O}_X$  has projective dimension exactly  $c = \mathrm{codim} X$ , by the Auslander–Buchsbaum formula

$$\mathrm{pd}_A M + \mathrm{depth}_A M = \mathrm{depth} A.$$

The resulting length- $c$  locally free resolutions provide the effaceability needed to show that every  $\theta_{\mathcal{F}}^i$  is an isomorphism.

There is also a geometric proof over an infinite field. A general linear projection gives a finite morphism

$$X \longrightarrow \mathbb{P}^n.$$

It is flat by miracle flatness, since  $X$  is Cohen–Macaulay and  $\mathbb{P}^n$  is regular of the same dimension. Duality can then be transported from projective space along this finite flat morphism.

## 10.5 Finite morphisms and exceptional pullback

Let  $f: X \rightarrow Y$  be finite and put

$$\mathcal{A} = f_* \mathcal{O}_X.$$

Under the equivalence between sheaves on  $X$  and  $\mathcal{A}$ -modules on  $Y$ , define

$$f^! \mathcal{G} = \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{A}, \mathcal{G}).$$

Locally, for  $A \rightarrow B$ , this is the  $B$ -module

$$\mathrm{Hom}_A(B, M), \quad (b \cdot \phi)(b') = \phi(bb').$$

**Proposition 10.6.** *The functor  $f^!$  is right adjoint to  $f_*$ :*

$$\mathrm{Hom}_X(\mathcal{F}, f^! \mathcal{G}) \simeq \mathrm{Hom}_Y(f_* \mathcal{F}, \mathcal{G}).$$

*Proof.* For a  $B$ -module  $N$  and an  $A$ -module  $M$ ,

$$\mathrm{Hom}_B(N, \mathrm{Hom}_A(B, M)) \simeq \mathrm{Hom}_A(N, M).$$

The correspondence sends  $\psi$  to  $n \mapsto \psi(n)(1)$ , with inverse

$$\phi \longmapsto [n \mapsto (b \mapsto \phi(bn))].$$

□

**Proposition 10.7.** *Suppose  $Y$  is projective and has dualizing sheaf  $\omega_Y$ . Then*

$$\omega_X = f^! \omega_Y$$

*is a dualizing sheaf on  $X$ . Its trace is the composite*

$$H^n(X, f^! \omega_Y) \simeq H^n(Y, f_* f^! \omega_Y) \longrightarrow H^n(Y, \omega_Y) \xrightarrow{\mathrm{tr}_Y} k.$$

*Proof.* The functor  $f_*$  is exact, and

$$H^n(X, \mathcal{F}) \simeq H^n(Y, f_* \mathcal{F}).$$

The adjunction identifies the duality pairing on  $X$  with the duality pairing for  $f_* \mathcal{F}$  on  $Y$ . □

If  $f$  is finite flat, then  $\mathcal{A}$  is locally free and  $f^!$  is exact. Derived adjunction gives

$$\mathrm{Ext}_X^i(\mathcal{F}, f^! \mathcal{G}) \simeq \mathrm{Ext}_Y^i(f_* \mathcal{F}, \mathcal{G}).$$

For an arbitrary finite morphism, the correct statement uses

$$Rf^! \mathcal{G} = R\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{A}, \mathcal{G}).$$

## 10.6 A concise introduction to derived categories

Let  $\mathcal{A}$  be an abelian category. A cochain complex is a sequence

$$\cdots \longrightarrow A^{i-1} \xrightarrow{d^{i-1}} A^i \xrightarrow{d^i} A^{i+1} \longrightarrow \cdots$$

with  $d^i d^{i-1} = 0$ . Its cohomology is

$$H^i(A^\bullet) = \ker d^i / \mathrm{im} d^{i-1}.$$

A chain map is a *quasi-isomorphism* if it induces isomorphisms on all cohomology objects.

### 10.6.1 Homotopy and localization

The homotopy category  $K(\mathcal{A})$  has complexes as objects and chain maps modulo chain homotopy as morphisms. The derived category  $D(\mathcal{A})$  is obtained by formally inverting all quasi-isomorphisms:

$$K(\mathcal{A}) \longrightarrow D(\mathcal{A}).$$

Morphisms may be represented by roofs

$$A^\bullet \xleftarrow{\simeq} C^\bullet \longrightarrow B^\bullet.$$

An acyclic complex becomes zero in  $D(\mathcal{A})$ , and the natural embedding

$$\mathcal{A} \hookrightarrow D(\mathcal{A})$$

is fully faithful.

The bounded derived category  $D^b(\mathcal{A})$  consists of objects whose cohomology vanishes outside a finite range. The categories  $D^+(\mathcal{A})$  and  $D^-(\mathcal{A})$  are defined using cohomology bounded below and above, respectively.

### 10.6.2 Triangles and cones

The shift is defined by

$$(A^\bullet[1])^i = A^{i+1}, \quad d_{A[1]}^i = -d_A^{i+1}.$$

Every chain map  $u: A^\bullet \rightarrow B^\bullet$  has a mapping cone, and

$$A^\bullet \longrightarrow B^\bullet \longrightarrow \text{Cone}(u) \longrightarrow A^\bullet[1]$$

is a distinguished triangle. Applying cohomology gives a long exact sequence. In particular, a short exact sequence in  $\mathcal{A}$  determines a distinguished triangle.

### 10.6.3 Derived functors

Let  $F: \mathcal{A} \rightarrow \mathcal{B}$  be left exact, and assume that  $\mathcal{A}$  has enough injectives. Replacing a bounded-below complex by an injective resolution defines

$$RF: D^+(\mathcal{A}) \longrightarrow D^+(\mathcal{B}), \quad R^i F = H^i \circ RF.$$

Examples include

$$R\Gamma(X, -), \quad Rf_*, \quad R\text{Hom}(A, -), \quad R\mathcal{H}om(\mathcal{F}, -).$$

Right exact functors are treated dually using projective or flat resolutions; this produces  $Lf^*$  and  $- \otimes^L -$ .

For objects  $A^\bullet, B^\bullet \in D(\mathcal{A})$ , define

$$\mathrm{Ext}^i(A^\bullet, B^\bullet) = \mathrm{Hom}_{D(\mathcal{A})}(A^\bullet, B^\bullet[i]).$$

For objects of  $\mathcal{A}$ , this recovers the classical Ext groups.

## 10.7 Spectral sequences

**Definition 10.8.** A cohomological spectral sequence consists of objects  $E_r^{p,q}$  and differentials

$$d_r^{p,q}: E_r^{p,q} \longrightarrow E_r^{p+r, q-r+1}$$

such that  $E_{r+1}^{p,q}$  is the cohomology at  $E_r^{p,q}$ . Under suitable boundedness, the pages stabilize to  $E_\infty^{p,q}$ , and these stable terms are the graded pieces of a filtration on the abutment:

$$E_\infty^{p,q} \simeq F^p H^{p+q} / F^{p+1} H^{p+q}.$$

We write

$$E_r^{p,q} \Longrightarrow H^{p+q}.$$

**Theorem 10.9** (Grothendieck spectral sequence). *Let*

$$\mathcal{A} \xrightarrow{F} \mathcal{B} \xrightarrow{G} \mathcal{C}$$

*be left exact functors between abelian categories with enough injectives. Suppose  $F$  sends injective objects to  $G$ -acyclic objects. Then*

$$E_2^{p,q} = R^p G(R^q F(A)) \Longrightarrow R^{p+q}(G \circ F)(A).$$

Taking

$$F = \mathcal{H}om(\mathcal{F}, -), \quad G = \Gamma(X, -)$$

gives the local-to-global Ext spectral sequence

$$E_2^{p,q} = H^p(X, \mathcal{E}xt^q(\mathcal{F}, \mathcal{G})) \Longrightarrow \mathrm{Ext}_X^{p+q}(\mathcal{F}, \mathcal{G}).$$

After a sufficiently positive twist, Serre vanishing kills all terms with  $p > 0$ , giving

$$\mathrm{Ext}_X^q(\mathcal{F}, \mathcal{G}(m)) \simeq H^0(X, \mathcal{E}xt^q(\mathcal{F}, \mathcal{G})(m)) \quad (m \gg 0).$$

## 10.8 Serre functors and Grothendieck–Verdier duality

**Definition 10.10.** Let  $\mathcal{D}$  be a  $k$ -linear triangulated category with finite-dimensional Hom spaces. A *Serre functor* is an exact autoequivalence  $S: \mathcal{D} \rightarrow \mathcal{D}$  equipped with functorial isomorphisms

$$\mathrm{Hom}_{\mathcal{D}}(A, B) \simeq \mathrm{Hom}_{\mathcal{D}}(B, S(A))^\vee.$$



**Theorem 10.11.** *If  $X$  is smooth and projective of dimension  $n$ , then*

$$S_X(-) = (-) \otimes \omega_X[n]$$

*is a Serre functor on  $D^b(\text{Coh } X)$ .*

Equivalently, for  $\mathcal{F}^\bullet, \mathcal{G}^\bullet \in D^b(\text{Coh } X)$ ,

$$\text{Ext}^i(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \simeq \text{Ext}^{n-i}(\mathcal{G}^\bullet, \mathcal{F}^\bullet \otimes \omega_X)^\vee.$$

The relative form is Grothendieck–Verdier duality. For a proper morphism  $f: X \rightarrow Y$ , the functor  $Rf_*$  admits a right adjoint  $f^!$ , and

$$Rf_* R\mathcal{H}om_X(\mathcal{F}^\bullet, f^! \mathcal{G}^\bullet) \simeq R\mathcal{H}om_Y(Rf_* \mathcal{F}^\bullet, \mathcal{G}^\bullet).$$

When  $X$  and  $Y$  are smooth over  $k$ , define

$$\mathbb{D}_X(-) = R\mathcal{H}om_X(-, \omega_X[\dim X]).$$

Then duality can be written succinctly as

$$Rf_* \mathbb{D}_X \simeq \mathbb{D}_Y Rf_*.$$

**Theorem 10.12** (Bondal–Orlov reconstruction). *Let  $X$  and  $Y$  be smooth projective varieties. If either  $\omega_X$  or  $\omega_X^\vee$  is ample, then an exact  $k$ -linear equivalence*

$$D^b(\text{Coh } X) \simeq D^b(\text{Coh } Y)$$

*implies  $X \simeq Y$ .*

Without the positivity hypothesis, nonisomorphic varieties can have equivalent derived categories; such examples are called Fourier–Mukai partners.

## 10.9 Applications

### 10.9.1 Curves

Let  $C$  be a smooth projective integral curve. If

$$\deg L > 2g(C) - 2,$$

then

$$H^1(C, L) \simeq H^0(C, \omega_C \otimes L^\vee)^\vee = 0,$$

because  $\omega_C \otimes L^\vee$  has negative degree.

If  $\deg L > 0$ , then

$$H^1(C, L \otimes \omega_C) \simeq H^0(C, L^\vee)^\vee = 0.$$

This is the curve case of Kodaira vanishing, and the argument works in every characteristic.

### 10.9.2 A curve on a surface

Let  $C \subseteq S$  be a smooth curve on a smooth projective surface. Tensoring

$$0 \longrightarrow \mathcal{O}_S(-C) \longrightarrow \mathcal{O}_S \longrightarrow \mathcal{O}_C \longrightarrow 0$$

by  $\omega_S(C)$  and using adjunction gives

$$0 \longrightarrow \omega_S \longrightarrow \omega_S(C) \longrightarrow \omega_C \longrightarrow 0.$$

If  $\mathcal{O}_S(C)$  is ample and  $\text{char } k = 0$ , then Kodaira vanishing and Serre duality give

$$H^1(S, \omega_S(C)) = H^2(S, \omega_S(C)) = 0.$$

Hence

$$H^1(C, \omega_C) \simeq H^2(S, \omega_S).$$

For connected  $C$  and  $S$ , both sides are naturally one-dimensional over  $k$ .

### 10.9.3 Varieties with trivial canonical bundle

If  $\omega_X \simeq \mathcal{O}_X$ , as for an abelian variety or a K3 surface, Serre duality becomes

$$\text{Ext}_X^i(\mathcal{F}, \mathcal{G}) \simeq \text{Ext}_X^{n-i}(\mathcal{G}, \mathcal{F})^\vee.$$

Equivalently, the Serre functor is the pure shift  $[n]$ . Thus  $D^b(\text{Coh } X)$  is an  $n$ -Calabi–Yau category in this sense.